The Social Costs of the Current Expected Credit Loss Approach: Evidence from Mortgage Lending*

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Abstract: I examine the impact of the current expected credit loss (CECL) approach on geographic disparities in mortgage credit access between low- and moderate-income (LMI) and non-LMI census tracts. Banks are subject to the Community Reinvestment Act (CRA), a key policy to reduce credit access inequality for borrowers in LMI census tracts. Using a triple-difference design, I find that following CECL adoption, banks reduce mortgage approval rates and increase mortgage interest rates in LMI census tracts, suggesting a decrease in mortgage credit supply by CECL-adopting banks in these areas. Non-bank lenders mitigate the reduction in bank credit. I show that following CECL adoption, non-bank lenders increase mortgage approval rates in LMI census tracts with greater exposure to CECL-adopting banks, but do not charge higher interest rates when extending mortgage credit in these areas. Findings highlight that the CECL approach has social costs by reducing banks' promotion of equitable credit access under the CRA. Results also inform the policy debate on extending CRA lending requirements to non-bank lenders.

Keywords: Current Expected Credit Loss Approach (CECL), Community Reinvestment Act (CRA), Mortgages, Banks, Non-banks, Real Effects
JEL classifications: G21, G23, G28, M41, M48

1. Introduction

Fair credit access for households is crucial to financial inclusion and social equity. The Community Reinvestment Act (CRA), enacted in 1977 in the United States, is the key law aimed at reducing disparity in credit access, requiring banks to meet the credit needs of the communities in which they operate and targeting borrowers in low- and moderate-income (LMI) census tracts. Regulators assess banks' compliance with the CRA periodically by evaluating their lending activities in LMI census tracts, defined as census tracts where median family income (MFI) is less than 80 percent of the MFI of the surrounding area, typically an MSA or a non-metropolitan area within the state. However, banks' compliance with the CRA is shaped by their incentives and capacity.

In response to the 2007/2009 financial crisis, the Financial Accounting Standards Board (FASB) introduced the current expected credit loss (CECL) model in 2016, changing how banks estimate and accrue provisions for credit losses fundamentally. The CECL approach replaced the incurred loss model (ILM), under which banks recognize credit losses only when they are incurred, probable, and reasonably estimable. In contrast, the CECL approach requires recognition of expected credit losses over the remaining life of loans at the end of each reporting period. Accounting Standard Update (ASU) 2016-13 required large public banks to adopt CECL at the beginning of their 2020 fiscal years, though it did not require small public and private banks to adopt it until 2023.

I examine how adoption of the CECL approach affects geographic disparities in mortgage credit access between LMI and non-LMI census tracts. I first examine whether CECL-adopting banks (CECL banks), subject to CRA lending requirements, reduce mortgage lending in LMI census tracts relative to non-LMI census tracts following CECL adoption. I then analyze whether non-bank lenders, not subject

to regulatory capital and CRA lending requirements, mitigate resulting increases in geographic disparities in mortgage credit access following CECL adoption.[1]

CECL significantly impacts banks' mortgage loans. Under the ILM, banks assess losses from homogeneous loans, such as mortgages, auto loans, and consumer credit, at the portfolio level, accruing provisions for losses expected over a short period–typically 12 to 18 months– following the reporting date (Ryan, 2019). Since mortgages' typical 30-year term far exceeds 12 to 18 months, CECL requires banks to record substantially higher loan loss provisions (LLPs) at origination. The increase in LLPs raises the amount of regulatory capital needed to support the loan, increasing the regulatory capital cost of mortgage origination. These costs are particularly elevated for mortgages in LMI census tracts because such mortgages have greater default rates (Saadi, 2020) and lower risk-adjusted interest rates (Cespedes et al., 2024) than those in non-LMI census tracts.[2]

I hypothesize that CECL decreases the quantity of mortgage credit supplied by CECL banks in LMI census tracts. Following CECL adoption, banks experience a trade-off between continuing to extend credit in LMI census tracts beyond the optimal level to meet CRA requirements and reducing credit in these areas.[3] Banks that fail to meet CRA lending requirements incur regulatory costs that include restrictions on mergers, acquisitions, and branch openings, and reputation harm. Extending credit in LMI census tracts beyond the optimal level incurs greater regulatory capital costs under CECL. Reducing credit in these areas avoids these costs, but risks violating CRA lending requirements.

[1] Following Avery et al. (2007), I define non-bank lenders as non-depository financial institutions that originate mortgage loans without affiliation with a traditional bank or savings institution.
[2] CECL also increased allowance on *existing* mortgages at adoption. However, banks adopting CECL in 2020 could elect regulatory capital transition relief that deferred the day-1 capital effect for two years and phased it in over the following three years.
[3] Optimal lending in LMI census tracts refers to the profit-maximizing level a bank would choose without CRA lending requirements.

I argue that the increased regulatory capital costs for CECL banks to originate mortgages in LMI census tracts are sufficiently high, leading such banks to reduce the quantity of mortgage credit in these areas. A reduction in the quantity of mortgage credit supplied by CECL banks, holding demand constant, should lead to higher interest rates. I thus expect that CECL banks increase mortgage interest rates in LMI census tracts following CECL adoption.

To test these hypotheses, I use a sample of conventional mortgage application records reported by banks under the Home Mortgage Disclosure Act (HMDA) from 2018 to 2022, which includes two years prior to and three years after CECL adoption.[4] My identification strategy uses a triple-difference estimation approach, comparing differences in changes in mortgage approval rates for CECL banks versus non-CECL banks across LMI versus non-LMI census tracts, before versus after CECL adoption in 2020. Focusing on mortgage approval decisions allows me to distinguish mortgage supply from mortgage demand (Acharya and Ryan, 2016; Dou et al., 2018; Loutskina and Strahan, 2009).

The empirical design includes bank-year and census tract-year fixed effects to control for time-varying bank characteristics and economic conditions in a census tract. I also include a CECL bank indicator interacted with census tract fixed effects to control for time-invariant disparities regarding how CECL and non-CECL banks allocate lending across census tracts. I further include county-bank fixed effects to control for each bank's lending activities across local mortgage markets. The design is robust to bank-year shocks that affect LMI and non-LMI lending similarly, to tract-year shocks such as the differential impact of the COVID-19 pandemic on LMI and non-LMI census tracts that affect CECL and non-CECL banks similarly, and persistent differences in bank-geography lending decisions.

Small public and private banks were exempt from CECL during the sample period, and thus non-CECL banks are smaller than CECL banks. To address concerns that differences in bank characteristics,

[4] The sample period begins with 2018 because variables, including loan interest rates, loan-to-value (LTV) ratio, and binned debt-to-income (DTI) ratio, became available from the HMDA dataset in that year.

such as size, drive results, I exclude CECL banks with total assets greater than 30 billion, and I apply entropy-balancing weights to achieve balance in the mean of bank characteristics. I then estimate an entropy-balanced-weighted triple-difference model. Consistent with my hypothesis, I find that following CECL adoption, CECL banks reduced mortgage approval rates in LMI census tracts by 1.4 percent more than non-CECL banks, representing an economically significant reduction of $126 million in mortgage credit from CECL banks to households in LMI census tracts.

Next, I examine changes in interest rates. Applying the same entropy-balanced-weighted triple-difference model, I find that following CECL adoption, CECL banks increased risk-adjusted mortgage interest rates for borrowers in LMI census tracts by 3 basis points more than non-CECL banks did. Cespedes et al. (2024) document that risk-adjusted mortgage interest rates are 2 basis points lower in LMI census tracts than in non-LMI census tracts using a sample before CECL adoption, and thus my estimate suggests that CECL banks eliminated the 2-basis-point interest rate discount in LMI census tracts following CECL adoption.

A central concern of my empirical strategy is that CECL adoption coincided with the onset of the COVID-19 pandemic. The estimates could be confounded if pandemic shocks affected CECL and non-CECL banks differently and also changed their lending to LMI and non-LMI census tracts. I address this concern in the following ways. First, I account for three major pandemic policies that might affect mortgage lending: the Coronavirus Aid, Relief, and Economic Security (CARES) Act mortgage forbearance and foreclosure protections, Federal Reserve purchases of agency mortgage-backed securities (MBS), and support for agency securitization. Using predetermined measures of banks' exposure to each channel, I allow these exposures to generate differential post-period changes in banks' lending gaps between LMI and non-LMI tracts. The estimates remain robust when the exposures are included separately and jointly. Second, I find no effect during the 2007/2009 financial crisis, a severe

economic downturn also accompanied by extensive policy intervention, or for federally insured or guaranteed mortgage loans and SBA 7(a) loans, for which government guarantees substantially reduce lenders' exposure to credit losses. Third, I obtain similar results around the 2023 CECL adoption by small public and private banks, which did not coincide with the pandemic. Fourth, the results are robust to restricting the sample to counties or census tracts in which both CECL and non-CECL banks operate, ensuring that they face similar local shocks. Finally, dynamic estimates extending the pre-period to 2016 show no differential trends before CECL adoption. Collectively, these analyses reduce the concern that the results reflect the pandemic or related policy interventions rather than CECL.

After showing that CECL banks decreased mortgage credit supply in LMI census tracts after CECL adoption, I examine whether non-bank lenders mitigated credit reduction. Unlike banks, non-bank lenders are not subject to the regulatory capital requirement, and they do not incur additional costs of regulatory capital for mortgage origination under the CECL approach. Buchak et al. (2018) show a rise in non-bank lenders in mortgage markets that banks find unprofitable due to regulatory constraints, facilitated by an originate-to-distribute (OTD) business model and government-sponsored enterprise (GSE) guarantees. When lending opportunities arise, non-bank lenders might increase credit in these areas, particularly those exposed more to CECL banks. Focusing on LMI census tracts, I thus hypothesize that non-bank lenders expanded the quantity of mortgage credit supplied in LMI census tracts with greater exposure to CECL banks following CECL adoption.

If borrowers left by CECL banks have lower credit quality than those served by non-bank lenders, non-bank lenders might charge higher risk premiums to them than banks do, since non-bank lenders are not subject to the CRA and have fewer incentives to serve LMI census tracts at favorable terms. However, higher risk-adjusted interest rates might decrease such borrowers' capacity to repay the debt, increasing the default probability and impacting profitability negatively. Conversely, if borrowers left by CECL

banks are of comparable or higher credit quality, controlling for demand, an increase in the quantity of mortgage credit supplied by non-bank lenders might lead to lower interest rates. Thus, it remains an empirical question whether non-bank lenders increase mortgage interest rates in LMI census tracts exposed more to CECL banks following CECL adoption.

To test these hypotheses, I use a sample of HMDA conventional mortgage application records from non-bank lenders. I use a triple-interaction estimator with a continuous exposure measure that examines whether, following CECL adoption, non-bank lenders increase mortgage approval rates more in LMI census tracts with greater exposure to CECL banks than in non-LMI census tracts. I measure a census tract's exposure to CECL banks by the amount of mortgage loans originated by CECL banks divided by the amount of mortgage loans originated by CECL banks, non-CECL banks, and non-bank lenders in a census tract from 2018 to 2019 (i.e., the pre-period). The specification includes lender-year, county-year, census tract, and lender-county fixed effects. The lower-order interaction accounts for post-period changes associated with a census tract's exposure to CECL banks common to LMI and non-LMI tracts, such as a greater shift toward remote lending in areas more reliant on CECL banks.

Consistent with the hypothesis, I find that following CECL adoption, increasing a census tract's exposure to CECL banks from the first quartile to the third quartile is associated with a 0.3 percent larger increase in approval rates in LMI census tracts than in non-LMI census tracts, representing an economically significant $256 million increase in mortgage credit from non-bank lenders to households in LMI census tracts.[5] I also examine changes in interest rates. Applying the same triple-interaction estimator, I find that non-bank lenders did not change their interest rates in LMI census tracts with greater exposure to CECL banks after CECL. As a placebo test, I re-estimate the specification using

[5] Loutskina and Strahan (2009) document that moving bank balance sheet liquidity from the first to third quartile increases mortgage approval rates by 0.6%. Dou et al. (2024) find that moving exposure to CFPB complaints from the first to third quartile increases peer banks' mortgage approval rates by 0.38%. The economic magnitude of my estimate is comparable to these studies.

government-guaranteed mortgage loans and find economically small and statistically insignificant effects on both approval rates and interest rates.

This paper makes incremental contributions to the literature on the CECL approach, non-bank lenders, and the CRA. I document that CECL banks reduce mortgage credit supply in LMI census tracts in comparison to non-CECL banks. Findings highlight that CECL imposes unintended social costs by weakening banks' role in mitigating the mortgage credit gap between LMI and non-LMI census tracts under the CRA. I also show that non-bank lenders responded to the credit reduction by increasing mortgage approval rates in LMI census tracts exposed more to CECL banks, without raising interest rates. By examining how CECL shapes mortgage credit disparities between LMI and non-LMI census tracts across types of lenders, this paper suggests implications for policymakers seeking to reduce homeownership gaps (Eggers, 2001), assessing the effects of financial reporting on social welfare (Ball, 2024). Findings further suggest that extending CRA lending requirements to non-bank lenders might be necessary to promote credit supply to LMI households that experience reduced access due to regulatory changes that affect banks.[6]

There are two caveats to this study. First, despite granular fixed effects, controls for major pandemic policies, and several robustness tests, since the COVID-19 pandemic was unprecedented and combined severe economic downturns with wide-ranging policy interventions, the paper cannot fully address unobserved pandemic shocks that affected CECL and non-CECL banks differently across LMI and non-LMI census tracts. Also, the findings on non-bank lenders are consistent with non-bank lenders mitigating the reduction in mortgage supply by CECL banks, although they cannot fully eliminate alternative explanations for differential non-bank growth during the pandemic. Second, the size

[6] Rapid growth of non-bank lenders in mortgage lending has raised calls to extend CRA requirements to these entities. For example, Federal Reserve chair Jerome Powell supports expansion of CRA to non-bank lenders (Kiernan and Ackerman, 2021); Massachusetts and Illinois law required non-bank lenders to comply with CRA requirements starting in 2024 (Goodman et al., 2023).

threshold improves comparability between CECL and non-CECL banks but limits external validity. I show that the results are robust to increasing the size threshold to $50 billion, extending the findings through mid-sized banks. The economic mechanism may apply to larger banks: CECL imposes greater capital costs on mortgage origination, while CRA imposes regulatory costs for decreasing mortgage supply in LMI census tracts. However, larger banks have more diversified portfolios and funding sources, greater access to securitization, and different capital and regulatory environments. The estimated magnitudes might not be extrapolated to larger banks. The effect of CECL on larger banks' lending in LMI communities and their contribution to aggregate mortgage credit remain open questions.

2. Related Literature and Contribution

This paper contributes to the literature on how loan loss provisioning and CECL adoption affect bank lending. Prior studies demonstrate that banks with timelier LLPs experience less reduction in loan growth during economic downturns (Beatty and Liao, 2011) and reduce risk-taking (Bushman and Williams, 2012). Recent research examines adoption of CECL and its effects on bank lending and credit supply. Basu et al. (2023) find that banks reduced credit supply and tightened loan terms due to a lack of guidance from standard-setters prior to CECL adoption. Yang (2025) studies banks' anticipatory response to CECL and finds that, at the bank-level, following CECL's approval, capital-constrained banks reduced residential loan growth and sold a greater share of newly originated mortgages. Chen et al. (2025) document that banks that adopted CECL reduced loan growth more during the COVID-19 recession than non-adopting banks did. Granja and Nagel (2025) (GN, henceforth) find that CECL's increased reserve requirements induced a moderate increase in interest rates of auto loans and personal unsecured loans, with the increase greater among borrowers with low credit quality at weakly-capitalized banks. Ballew et al. (2022) show that, of banks that participated in the Paycheck Protection Program,

non-CECL banks increased risk-taking more than CECL banks did. Some theory papers suggest that CECL adoption reduces lending procyclicality (e.g., Chae et al., 2018; Huber, 2022). Mahieux et al. (2023) show that the expected loss model encourages credit supply and improves financial stability when regulators tailor regulatory capital to incorporate information about credit losses. However, Abad and Suarez (2018) show that expected loss models increase lending procyclicality.

Most related to the current paper, GN examine auto and personal unsecured loans, finding that i) CECL banks increase interest rates, with stronger effects for higher-credit-score individuals; ii) among weakly capitalized CECL banks, interest rate increases are largest among borrowers with the lowest credit scores (FICO 550-599); and iii) CECL banks reduce loan amounts among borrowers with low to fair credit scores (FICO 650-699), but not for those with other credit scores. This paper makes the following contributions that are incremental to GN. First, I examine the effect of CECL adoption on the gap in mortgage credit supply between LMI and non-LMI census tracts. Mortgages are economically important given the role they play in household wealth accumulation and homeownership (Eggers, 2001). Larger volumes and longer maturities of mortgage loans, relative to auto and unsecured consumer credit, make mortgage lending a critical context in which to assess the effects of CECL. Banks rely heavily on mortgage loans to satisfy CRA lending requirements.[7] Focusing on the LMI and non-LMI mortgage credit gap allows me to analyze how the CECL accounting standard interacts with CRA lending requirements to shape geographic disparities in banks' credit supply. I show that CECL banks reduce the *flow* (i.e., originations) of mortgage credit to LMI census tracts, whereas GN document reductions in the *level* (i.e., amounts) of consumer loans among borrowers with FICO scores between 650 and 699. I find that CECL banks increase mortgage interest rates more in LMI census tracts in comparison to non-LMI census tracts, whereas GN find larger rate increases for higher-credit-score individuals. I examine

[7] Bank regulators focus on mortgage and small business lending quantities when assessing banks' compliance with CRA (Agarwal et al., 2012). A detailed discussion of CRA appears in Section 3.2.

how non-bank lenders, not subject to CRA lending requirements, respond to lending slack from CECL adoption. These results provide new evidence on how CECL affects geographic disparities in credit supply across types of lenders, contributing to ongoing policy debates on equitable credit access and the potential extension of CRA requirements to non-bank institutions. My study also differs from Yang (2025) by providing mortgage application-level evidence on how realized CECL adoption affects mortgage approval and pricing across LMI and non-LMI census tracts.

This paper also contributes to the literature on the rise of non-bank lenders. Buchak et al. (2018) find that the market share of mortgage originations by non-bank lenders increased from less than 30 percent in 2007 to more than 50 percent in 2015 due to regulatory constraints banks experienced post-crisis. Fuster et al. (2019) and Allen et al. (2023) document that FinTech lenders responded more elastically to mortgage demand shocks, including those arising from natural disasters, than banks did. Gete and Reher (2021) find that bank liquidity coverage ratio requirements contribute to the growth of non-bank lenders in the mortgage market share. Begley and Srinivasan (2022) find that small banks exploited regulations that affected the largest banks disproportionately and relied on lending on the balance sheet to persist in the mortgage market during the rise of non-bank lenders.

Some papers document that the shift of lending to non-bank lenders, again not subject to CRA lending requirements, harms the credit access of the LMI population. Cespedes et al. (2024) find that CRA widens disparities in credit access in underserved regions due to non-bank lenders' growth. Chu et al. (2023) document higher denial rates for small mortgage applications in areas exposed more to FinTech and shadow banks, despite similar application quality and local economic trends, and that those increased denial rates contribute to lower homeownership among less affluent households. Non-bank lenders also pose risks to financial system stability due to their limited balance sheet capacity (Buchak et al., 2024), loan risk-taking (Irani et al., 2021), and inadequate risk-based capital (Kim et al., 2022).

The current paper evidences that non-bank lenders improve economic welfare by supplying credit to LMI households that otherwise are unable to access it from banks due to the CECL accounting standard.

This paper relates to CRA literature. Extant literature examines CRA's effects on mortgage lending in LMI neighborhoods (Avery et al., 2003; Bhutta, 2011; Begley and Purnanandam, 2021; Conway et al., 2023; Ding and Nakamura, 2021; Lee and Bostic, 2020), though none examines its interaction with accounting standard changes. Regarding loan performance, Agarwal et al. (2012) and Saadi (2020) find that CRA increases mortgage default rates. Cespedes et al. (2024) show that risk-adjusted mortgage rates are lower in LMI census tracts with an MFI just below the 80 percent threshold. However, evidence suggests that banks, for example, limit their size (Cespedes et al., 2024) and close branches in LMI neighborhoods (Cespedes et al., 2023) to avoid CRA regulations. This paper complements this literature by showing that CRA is less effective at requiring banks to promote fair credit access when they experience regulatory changes, such as adoption of the CECL accounting standard.

3. Institutional Background and Hypotheses

3.1 Current Expected Credit Loss Approach

The primary difference between CECL and ILM is the timing of recognizing credit losses. Under ILM, financial institutions recognize credit losses only when they become incurred, probable, and reasonably estimable. Banks assess losses from homogeneous loans, such as mortgages, auto loans, and consumer credit loans, at the portfolio level, in an amount equal to losses expected over a short time horizon, typically 12 to 18 months following the reporting date (Ryan, 2019). CECL eliminates ILM's "probable" condition for recognizing credit losses, requiring financial institutions to recognize expected credit losses at the time of loan origination, equivalent to the current expected credit losses on loans over

their lifetime. CECL also relaxes the "estimable" condition under ILM. Banks are required to incorporate a broader range of information, including historical data, current economic conditions, and reasonable and supportable forecasts, into estimations of expected credit losses during the remaining life of a loan. CECL increases the cost of originating mortgage loans, since banks record higher LLPs at the time of origination, requiring more capital.[8]

Implementation of CECL follows several steps. In June 2016, the FASB issued ASU 2016-13, introducing CECL. Large public entities are required to adopt CECL for fiscal years beginning after December 15, 2019. Smaller reporting companies (SRCs) are required to adopt CECL for fiscal years beginning after December 15, 2022.[9] In 2018, FASB and bank regulators released guidance and clarifications to facilitate adoption of CECL. Banks began developing internal models and testing the new methodology, but the onset of the COVID-19 pandemic led to further delays. The CARES Act allowed firms to defer CECL adoption until the earlier of the end of the national emergency or December 31, 2020.

[8] Consider this example. A bank originates a portfolio of 1,000 mortgages, each with a principal balance of 200,000, to borrowers with credit scores of 750. The mortgages are 30-year, fully amortizing loans with a fixed interest rate of 4%. Assume a discount rate of 0%. Under CECL, the bank estimates that, at origination, *each* loan has a lifetime probability of default of 5% and a loss given default of 25%. The bank thus recognizes an LLP of \$2,500 *per* loan (\$200,000 × 5% × 25%), resulting in a total LLP of \$2,500,000 for the 1000 loans at origination. Under ILM, the bank monitors the *portfolio* for signs of credit losses incurred, probable, and reasonably estimable. Suppose the bank reasonably estimates that 1% of the loans will become delinquent for more than 60 days within one year of origination, and that the historical loss rate for such delinquent loans is 25%. The bank records an LLP of \$500,000 (1,000 × 1% × \$200,000 × 25%) for the portfolio at origination.

[9] A public company is an SRC if (i) it has a public float of less than \$250 million or (ii) it has less than \$100 million in annual revenues and either no public float or a public float of less than \$700 million. For more details, see https://www.sec.gov/education/smallbusiness/goingpublic/SRC.

3.2 The Community Reinvestment Act

3.2.1 Background

The CRA is a U.S. federal law enacted in 1977 that applies to all federally insured depository institutions, but it excludes credit unions and non-bank lenders. The CRA's purpose is to encourage federally insured depository institutions, such as banks and savings associations, to meet the credit needs of communities in which they operate, particularly LMI neighborhoods and households, and to promote fair access to credit.

Banks' compliance with CRA is assessed periodically and enforced by federal banking regulators through a comprehensive CRA examination, which includes three tests–lending, investment, and service. Among these, the lending test is most important, focusing on mortgages and small business lending (Agarwal et al., 2012). It assesses loan quantities and examines geographic distribution of lending to LMI census tracts within a bank's assessment areas.[10] Within each assessment area, lending in census tracts in which the MFI is below 80 percent of the MFI for the surrounding area–an MSA or a non-metropolitan area within the state–satisfies CRA lending requirements. Following a CRA examination, banks receive a public rating that reflects their degree of compliance. Institutions with unsatisfactory ratings are penalized, including denial of branch openings and merger applications, and reputational damage.[11]

[10] A bank's assessment area is defined as the region in which it operates branches and deposit-taking ATMs, or has significant lending activity.

[11] For example, Chen et al. (2023) find a decline in deposit growth after release of a downward CRA rating, and Dou and Zou (2019) find that banks originate under-performing commercial loans in LMI areas due to public pressure.

3.2.2 Contemporary Policy Development

In 2023, federal banking regulators issued rules to modernize CRA, scheduled to go into effect during 2026. The most significant revisions include introduction of more transparent evaluation metrics and an expanded definition of banks' assessment areas.[12] The updated rules establish clearer benchmarks, facilitating greater consistency during CRA evaluations across institutions. The new metrics and benchmarks are more stringent, making it more difficult for banks to receive the highest ratings (Ackerman, 2023). Whereas CRA assessments were based on the geographic areas in which banks maintained physical branches and accepted deposits, the new framework expands coverage to include areas in which banks engage in lending and deposit-taking activity, regardless of branch presence, addressing the growing prominence of digital banking.

The revised rules, however, never took effect. Industry groups and banks argue that the new regulations are overly complicated and impose significant compliance costs (Tokar, 2025). Critics warn of potential unintended consequences, such as banks reducing services or closing branches in LMI areas to avoid CRA assessments. Federal banking regulators announced in March 2025 that they would rescind implementation of the new CRA rules. They subsequently issued a joint proposal in July 2025 to replace it with regulations substantively equivalent to the preexisting CRA framework.[13]

State policy has moved in a different direction. Several states, such as New York, Massachusetts, and Illinois, have taken steps to expand CRA requirements to non-bank lenders, particularly non-bank mortgage companies, in response to their growth in the mortgage market (Goodman et al., 2023). In January 2026, for example, the New York State Department of Financial Services finalized a regulation

[12] For details on the updated CRA rules, see the summary provided by the Federal Reserve Board at https://www.federalreserve.gov/consumerscommunities/community-reinvestment-act-final-rule.htm.
[13] See the Federal Reserve Board, FDIC, and OCC joint proposal to rescind the 2023 CRA rule, https://www.federalreserve.gov/newsevents/pressreleases/bcreg20250716a.htm.

that requires non-bank mortgage lenders to demonstrate support for equitable access to home loans in the communities they serve as an expansion of the CRA. The regulation became effective on July 7, 2026.[14] Such state-level initiatives reflect policy efforts to modernize CRA requirements to adapt to evolving financial intermediation.

3.3 Hypotheses

3.3.1 Mortgage Lending in LMI Census Tracts by Banks

Following CECL adoption, banks might continue to extend credit in LMI census tracts to satisfy CRA lending requirements, avoiding regulatory costs. However, lending to LMI census tracts is costly. Cespedes et al. (2024) document that risk-adjusted mortgage rates are 2 basis points lower in census tracts with an MFI just below the 80 percent threshold, in comparison to those just above, suggesting that CRA regulations compress profit margins on loans to desirable and low-risk LMI neighborhoods. Saadi (2020) documents higher mortgage default rates in LMI neighborhoods than in non-LMI neighborhoods. CECL raises costs of originating new mortgages by requiring more capital, and the increase in costs is higher in LMI census tracts than in non-LMI census tracts due to lower risk-adjusted returns and higher default rates. CECL also increased allowance on existing mortgages at adoption. However, banks adopting CECL in 2020 could elect regulatory capital transition relief that deferred the day-1 capital effect for two years and phased it in over the following three years. To the extent that banks had anticipated the future impact on capital, the increase in reserves for existing mortgages might constrain their lending, but such effect would likely have appeared gradually.

[14] For details, see https://www.dfs.ny.gov/reports_and_publications/press_releases/pr20260108.

Following CECL adoption, banks experience a trade-off between extending credit in LMI census tracts beyond the optimal level, as required by CRA, and reducing lending in these areas. Extending credit in this case requires higher regulatory capital costs but avoids CRA non-compliance penalties. Withdrawing from these areas decreases regulatory capital costs, but risks violating CRA lending requirements. If the increase in regulatory capital costs of originating mortgages in these areas is sufficiently large, CECL banks reduce mortgage credit supply in LMI tracts. However, if they operate near the threshold of non-compliance with CRA and if non-compliance costs are sufficiently high, they might maintain lending in such areas. I expect that the increase in regulatory capital costs of lending in LMI census tracts dominates, leading CECL banks to reduce the quantity of mortgage credit in these areas.

Controlling for mortgage credit demand, a reduction in the mortgage quantity supplied by CECL banks in LMI tracts should, in equilibrium, lead to higher mortgage interest rates. Thus:

H1A: *Following CECL adoption, CECL banks decrease the quantity of mortgage credit in LMI census tracts.*

H1B: *Following CECL adoption, CECL banks increase mortgage interest rates in LMI census tracts.*

3.3.2 Mortgage Lending in LMI Census Tracts by Non-bank Lenders

Unlike banks, non-bank lenders are not subject to regulatory capital requirements. Recent literature documents a rise of non-bank lenders in markets in which banks find it unprofitable due to regulatory constraints (Buchak et al., 2018). Non-bank lenders do not incur additional regulatory capital costs in mortgage origination under CECL. When lending opportunities arise, non-bank lenders are likely to extend more credit in LMI census tracts. Banks also retain a substantial share of mortgage loans on their balance sheets, but non-bank lenders rely nearly exclusively on securitization and the OTD

model to finance originations (Buchak et al., 2018).[15] This provides non-bank lenders with a financing advantage over banks, allowing them to fill a lending gap. Since LMI census tracts with greater exposure to CECL banks are more likely to experience a decline in mortgage credit due to CECL adoption, non-bank lenders identify the most lending opportunities in these areas. By focusing on LMI census tracts, non-bank lenders increase the quantity of mortgage credit supplied in census tracts exposed more to CECL banks.

Several factors suggest that non-bank lenders might not increase mortgage lending in these areas. Tomy (2019) documents that higher LLP deters entry into local banking markets. Dou et al. (2024) document that banks change their lending behaviors by learning from consumers' complaints about peers. If non-bank lenders observe an increase in CECL banks' LLPs and withdrawal from LMI neighborhoods, they might also reduce lending in LMI neighborhoods. I suggest that non-bank lenders increase the quantity of mortgage credit in census tracts exposed more to CECL banks.

If borrowers left by CECL banks are of lower credit quality than those that non-bank lenders typically serve, non-bank lenders might charge higher risk-adjusted interest rates to compensate for increased default risk, especially since they are not subject to CRA lending requirements. However, higher interest rates might decrease borrowers' ability to repay, raising default probabilities and reducing loan profitability. Conversely, if borrowers served previously by CECL banks are of comparable or higher credit quality, controlling for demand, an increase in the quantity of mortgage credit supplied might lead to a decrease in interest rates. Non-bank lenders also have low funding costs due to their OTD business model, and thus they might not increase risk-adjusted mortgage interest rates in LMI census tracts that are exposed more to banks' CECL adoption. Overall, it remains an empirical question whether non-bank lenders increase mortgage interest rates in these areas following CECL adoption. Thus:

[15] Banks retain 40% to 50% of mortgage loans on their balance sheets (Buchak et al., 2018, 2024; Dou et al., 2018).

H2A: *Following CECL adoption, non-bank lenders increase the quantity of mortgage credit in LMI census tracts with greater exposure to CECL banks.*

H2B: *Following CECL adoption, non-bank lenders may or may not increase mortgage interest rates in LMI census tracts with greater exposure to CECL banks.*

4. Data and Sample

4.1 Sample

My primary data source is mortgage application records reported under HMDA, which requires nearly all mortgage lenders to report detailed information on the applications they receive, and whether they originate the loan. The HMDA dataset provides application outcomes, applicant characteristics, loan characteristics, and the lender's identity. Since 2018, additional information, including interest rates, LTV ratios, and binned DTI ratios, is also available. The dataset also provides the census tract location of a property.

Sample construction begins with conventional mortgage loan applications to banks from the HMDA dataset for 2018 to 2022. Consistent with Dou et al. (2024), I apply the following selection criteria: (1) loans with an outcome of either approved or denied, and (2) a loan amount greater than $1000, yielding an initial sample of 16,636,776 applications from 3,094 unique banks across 72,786 census tracts.[16] I require valid values for borrower (i.e., gender, race, and income) and loan (i.e.,

[16] The 2020 Census survey includes redefinitions of census tracts from the 2010 Census survey. Some census tracts were split into smaller ones due to population growth, and some merged due to population declines. The HMDA dataset uses 2020 census tracts for data from 2022 onward, and 2010 census tracts for data from 2018 to 2021. I use the crosswalk provided by IPUMS NHGIS to map 2020 census tracts to 2010 census tracts. The crosswalk is available at https://www.nhgis.org/geographic-crosswalks.

conforming status, occupancy type, purpose, census tract location, LTV ratio, and binned DTI ratio) characteristics. The filters reduced the sample size to 12,288,009 loan applications and 1,476 banks.

I obtain financial data for commercial banks from the Report of Condition and Income (Call Report), and I link HMDA bank identifiers to Call Report data using the HMDA Reporter Panel. Observations with missing bank-level control variables are excluded from the analyses. Following Chen et al. (2025), I exclude banks that adopted CECL between 2021 and 2022. As SRCs are exempt from CECL adoption in the sample period, non-CECL banks are much smaller than CECL banks. To balance CECL and non-CECL banks based on size, I exclude applications from CECL banks with total assets that exceed $30 billion in 2019, the last pre-CECL year.[17] Table 1, Panel A is a summary of the sample selection procedure for mortgage applications submitted to banks. The final sample contains 5,402,941 mortgage applications from 129 CECL banks and 1,202 non-CECL banks in 71,582 census tracts.

Panel B of Table 1 outlines the sample selection procedure for mortgage loan applications submitted to non-bank lenders. Following Avery et al. (2007), I define such lenders as non-depository financial institutions that originate mortgage loans without affiliation with a traditional bank or savings institution. Selection criteria applied to non-bank lenders' applications are similar to those used for banks. I also require non-missing values for *CECL Share*, the tract-level exposure measure used in the non-bank analysis. The final sample contains 17,528,179 mortgage applications from 1,021 non-bank lenders in 55,607 census tracts.

[Insert Table 1 near here]

[17] Under the OCC's supervisory convention, banks with total assets above $50 billion are classified as large banks, those with $10 to $50 billion are mid-sized banks, and those under $10 billion are small community banks. Therefore, the sample primarily includes CECL banks in the lower mid-sized and small range. Results are robust to increasing the maximum size to $50 billion.

4.2 Summary Statistics

I first analyze the effects of CECL adoption on bank mortgage lending in LMI and non-LMI census tracts using mortgage applications from banks and subsequently examine mortgage lending by non-bank lenders. Table 2 reports summary statistics for mortgage applications submitted to both lender types by LMI census tracts, non-LMI census tracts, and all census tracts. All continuous variables are winsorized at the 1 percent level.

Panel A reports statistics for mortgage applications submitted to banks, showing that 90.3 percent of mortgage applications submitted to banks were approved. The approval rate for applications in LMI census tracts was 87.1 percent, in comparison to 90.8 percent for applications in non-LMI census tracts. The average borrower in non-LMI census tracts had a $Log(Income)$ of 11.584 and an LTV of 71.093 percent, in comparison to an average $Log(Income)$ of 11.276 and an LTV of 74.688 percent in LMI census tracts. Forty-one-and-a-half percent of borrowers in non-LMI census tracts had a DTI ratio above 36 percent, lower than 46.4 percent in LMI census tracts. Borrowers in LMI census tracts also paid higher interest rates than those in non-LMI census tracts (4.016 percent versus 3.743 percent in non-LMI census tracts).

Panel B reports statistics for mortgage applications submitted to non-bank lenders, which also had an overall approval rate of 90.5 percent. For non-bank lenders, the approval rate was also lower in LMI census tracts (88.5 percent) than in non-LMI census tracts (90.8 percent). The average borrower in LMI census tracts had a *Log(Income)* of 11.315 and an *LTV* of 74.688 percent, in comparison to a *Log(Income)* of 11.559 and an *LTV* of 72.156 percent. Similarly, 48.3 percent of borrowers in non-LMI census tracts had a DTI ratio above 36 percent, lower than 54.3 percent in LMI census tracts. Borrowers at non-bank lenders in LMI census tracts also paid higher interest rates than those in non-LMI census tracts (3.785 percent versus 3.611 percent in non-LMI census tracts).

[Insert Table 2 near here]

Tests of H1A and H1B rely on mortgage lending from non-CECL banks as a control group. To mitigate concerns regarding observable differences, particularly in bank size, between CECL and non-CECL banks, I use entropy balancing to achieve covariate balance of the means of several bank characteristics, including *Log(Size)*, *Deposit Ratio*, *Tier-1 Capital Ratio*, *Residential R/E Loan Ratio*, *C&I Loan Ratio*, and *NPL Ratio*.

Panel A of Table 3 reports summary statistics for bank characteristics. Panel B reports entropy-balancing statistics for bank characteristics, showing mean characteristics of CECL and non-CECL banks both before and after balancing. Before applying entropy balancing, CECL and non-CECL banks are different in characteristics except for *NPL Ratio*. After balancing, the panel demonstrates that entropy balancing achieves a high degree of covariate balance of mean characteristics, ensuring comparability between treatment and control groups.

[Insert Table 3 near here]

5. Effect on Bank Mortgage Lending

5.1 Empirical Design and Results

I first test H1A by analyzing whether CECL banks reduced mortgage approval rates in LMI census tracts. I then test H1B by examining whether CECL banks increased interest rates for originated mortgages in these areas. For both tests, I restrict the sample to mortgages at banks, excluding those at non-bank lenders, because banks are subject to similar regulatory and CRA lending requirements.

To test H1A, I use triple-difference estimation, comparing differences in changes in mortgage approval rates by CECL banks versus non-CECL banks in LMI versus non-LMI census tracts, both before and after CECL adoption. I estimate an entropy-balanced-weighted linear probability triple-difference model using mortgage applications submitted to CECL and non-CECL banks:

$$I(Approved)_{i,b,c,s,t} = \alpha + \beta_1 CECL_b \times CRA_c \times Post_t + \Gamma' X_{i,t} + \delta_{b,t} + \eta_{c,t} + \phi_{CECL_b,c} + \xi_{b,s} + \epsilon_{i,b,c,s,t}. \quad (1)$$

where i, b, c, s, and t index loans, banks, census tracts, counties, and years, respectively. The dependent variable, $I(Approved)_{i,b,c,s,t}$, equals 1 if bank b approved loan application i in census tract c during year t, and zero otherwise. Focusing on approval decisions helps distinguish mortgage supply from demand (Loutskina and Strahan, 2009; Acharya and Ryan, 2016; Dou et al., 2018). $CECL_{i,b}$ equals 1 if a bank adopted CECL during 2020, and zero otherwise. CRA_c equals 1 for LMI census tracts, and zero otherwise, and $Post_t$ equals 1 beginning in 2020, and zero for prior years.

The empirical strategy assumes that CECL imposes a larger provisioning on mortgages originated in LMI census tracts than on those originated in non-LMI census tracts because mortgages in LMI census tracts have higher default risk. To provide supporting evidence for this premise, I use quarterly Call Report data from 2018–2019 and regress the bank's allowance for loan losses (ALL) on residential real estate loans, divided by total assets, on its *LMI Market Exposure* and bank characteristics. I define *LMI Market Exposure* as the amount of mortgages originated in LMI census tracts during the pre-period divided by the total amount of mortgage originations during the same period.[18] The results, reported in Online Appendix OA.1, indicate that banks with greater pre-period exposure to LMI census tracts had higher ALL on residential real estate loans ratio. The coefficient estimate implies that an interquartile increase in the pre-period *LMI Market Exposure* increases ALL on residential real estate

[18] Summary statistics for *LMI Market Exposure* appear in Appendix B.

loans ratio by 0.016 percent, representing 16 percent of the sample median. Moreover, since regulators evaluate the geographic distribution of lending in LMI census tracts in assessing CRA compliance, the geographic comparison also maps directly to the geographic dimension of CRA evaluation.

I control for loan and borrower characteristics in vector $X_{i,t}$. Loan characteristics include *Log(Loan Amount)*, the natural logarithm of the loan amount; *Owner Occupied*, an indicator that equals 1 if the property was owner-occupied as a principal residence, and zero otherwise; *Loan Purpose*, an indicator that equals 1 for home purchase loans, and zero for refinancing; and *Conforming*, an indicator that equals 1 if the loan amount was below the conforming loan limit, and zero otherwise. I also control for borrower characteristics, including *Log(Income)*, the natural logarithm of applicant income; *LTV*, the LTV ratio of the applicant; *I(DTI >36%)*, an indicator that equals 1 if an applicant's DTI ratio was greater than 36 percent, and zero otherwise; *Female*, an indicator that equals 1 if an applicant's gender was female, and zero otherwise; and *White*, an indicator that equals 1 if an applicant's race was White, and zero otherwise.

The model includes bank-year and census tract-year fixed effects to control for time-varying bank characteristics and economic conditions in a census tract, including house prices and pandemic severity.[19] I also include a CECL bank indicator interacted with census tract fixed effects (i.e., CECL bank-census tract fixed effects) to control for time-invariant disparities regarding how CECL and non-CECL banks allocate lending across census tracts. I further include county-bank fixed effects to control

[19] A fully saturated model would include bank-tract fixed effects. However, the sample has about 5.4 million observations but more than 1.1 million bank-tract pairs, so there are fewer than 5 observations per bank-tract cell on average. The effective support is narrower because a bank-tract pair must provide usable within-pair variation after other fixed effects are absorbed, and only pairs that persist across pre- and post-periods provide meaningful variation. With so little within-cell variation, the within-bank-tract specification has very low power, consistent with the discussion in Breuer and deHaan (2024) and Correia (2015). Therefore, the design uses bank-county fixed effects.

for time-invariant differences in each bank's activities across local mortgage markets. I cluster standard errors at the bank level.

The triple-difference estimator is equivalent to the difference between two difference-in-differences (DiD) estimators. The first DiD estimator compares (i) the change in mortgage approval rates in LMI census tracts by CECL banks (group A) before and after CECL adoption and (ii) the change in mortgage approval rates in non-LMI census tracts by CECL banks (group B) before and after CECL adoption. The second DiD estimator compares (i) the change in mortgage approval rates in LMI census tracts by non-CECL banks (group C) before and after CECL adoption and (ii) the change in mortgage approval rates in non-LMI census tracts by non-CECL banks (group D) before and after CECL adoption. The triple-difference is the net difference between the two DiD estimates. The identifying assumption is that, without CECL, the relative change in outcomes between groups A and B would have trended in the same way as the relative change between groups C and D. Thus, the design addresses bank-year shocks that affect LMI and non-LMI tracts similarly, tract-year shocks, including divergences in economic trends between LMI and non-LMI tracts, that affect CECL and non-CECL banks similarly, and time-invariant differences in bank-geography lending patterns.

Since SRCs were exempted from CECL in the sample period, non-CECL banks are significantly smaller than CECL banks. To mitigate concerns that results are driven by differences in bank characteristics such as size, I apply entropy balancing weights to balance the mean of bank characteristics, including *Log(Size)*, *Deposit Ratio*, *Tier-1 Capital Ratio*, *Residential R/E Loan Ratio*, *C&I Loan Ratio*, and *NPL Ratio*, and I estimate the entropy-balanced-weighted linear probability triple-difference model.

Columns (1) and (2) of Table 4 report results. Both columns include bank-year, CECL bank-census tract, and census-tract fixed effects. Column (2) adds bank-county fixed effects. The coefficient for the interaction term $CECL \times LMI \times Post$ is negative and statistically significant in both columns.

The coefficient in Column (2) suggests that mortgage approval rates by CECL banks in LMI census tracts decreased by 1.4 percent relative to the change in approval rates by non-CECL banks in LMI census tracts after CECL adoption. This decrease corresponds to approximately 2 percent of the unconditional approval rates of 87.1 percent by CECL banks in LMI census tracts. The total mortgage application volume in LMI census tracts at CECL banks in the sample in 2019 was 9 billion, and thus the 1.4 percent decrease in mortgage approval rates represents an economically significant 126 million reduction in credit supply from CECL banks to households in LMI census tracts.

To test H1B, I focus on mortgages originated by banks and apply the same estimation approach. I replace the dependent variable in Eq. (1) with *Interest Rate*. Columns (3) and (4) of Table 4 report results. In Column (4), the positive and significant coefficient for $CECL \times LMI \times Post$ suggests that CECL banks increased interest rates in LMI census tracts by 3 basis points relative to the change at non-CECL banks, corresponding to approximately 2.8 percent of the standard deviation of the interest rate of mortgages originated by banks. Cespedes et al. (2024) document that risk-adjusted mortgage interest rates are 2 basis points lower in LMI census tracts than in non-LMI census tracts using a sample before CECL adoption, and thus my estimate suggests that CECL banks eliminated the interest rate discount in LMI census tracts following CECL adoption.

[Insert Table 4 near here]

To assess pre-trends, I estimate a modified version of Eq. (1), replacing $Post$ with year indicators for each year in the triple interaction term $CECL \times LMI \times Post$, where the dependent variables are *I(Approved)* and *Interest Rate*. Panel A of Figure 1 plots the coefficient estimates relative to 2018 and their 90 percent confidence intervals for approval rates. The plot suggests that approval rates did not

trend differently pre-CECL, and coefficients for 2020, 2021, and 2022 are negative and statistically significant.

Panel B of Figure 1 plots coefficient estimates relative to 2018 and their 90 percent confidence intervals for interest rates. The plot suggests that CECL banks increased risk-adjusted interest rates in LMI census tracts beginning with 2021.

[Insert Figure 1 near here]

To further assess pre-trends, I extend the analysis on approval rates to 2016 and estimate the same dynamic specification using 2016 as the base year.[20] Figure 2 plots the coefficients relative to 2016 and shows no differential trend before CECL adoption.

[Insert Figure 2 near here]

When reducing mortgage credit supply, CECL banks might tighten underwriting standards by lending to higher-credit-quality borrowers in LMI census tracts. As a supplemental analysis, in Online Appendix OA.2, I focus on mortgages approved by banks and replace the dependent variable in Eq. (1) with *I(High Risk)*, an indicator equals 1 if a borrower had both a high LTV ratio (i.e., above 90 percent) and a high DTI ratio (i.e., above 36 percent). Table OA.2 suggests that mortgages originated by CECL banks in LMI census tracts were 1.1 percent less likely to have a high-risk borrower, suggesting that following CECL adoption, banks tightened underwriting standards in LMI census tracts. Figure OA.2 illustrates dynamic effects of CECL adoption on banks' underwriting standards between non-LMI and LMI census tracts, showing no pre-trend. The post-period coefficients indicate CECL banks were less

[20] LTV ratios, DTI ratios, and interest rates are unavailable in the public HMDA data before 2018. Accordingly, the extended analysis omits the LTV and DTI controls and examines approval rates only.

likely to lend to a high-risk borrower in LMI census tracts beginning from the initial year of CECL adoption.

Results reported in this section suggest that CECL banks withdrew mortgage credit supply to LMI census tracts after CECL adoption. They decreased the *flow* (i.e., originations) of mortgage credit into these tracts. Consistent with a contraction in credit supply, CECL banks increased the risk-adjusted interest rates for borrowers in LMI census tracts. CECL banks also tightened their underwriting standards in these areas after CECL adoption.

5.2 Identification and Alternative Explanations

5.2.1 Pandemic Policies and Correlated Shocks

A central concern is that CECL adoption coincided with the onset of the COVID-19 pandemic. Pandemic-related shocks may have affected CECL and non-CECL banks differently across LMI and non-LMI census tracts. I examine three channels through which such differential effects could arise: the CARES Act mortgage forbearance and foreclosure moratoria, the Federal Reserve's large-scale purchases of agency MBS, and support for agency securitization. For each channel, I construct a predetermined bank-level exposure measure using 2018–2019 data and augment Eq. (1) with the exposure measure interacted with *CRA* and with *CRA* × *Post*. These terms allow banks with different pre-pandemic exposures to have different pre-period lending gaps between LMI and non-LMI census tracts and different changes in those gaps in the post period.[21]

The CARES Act forbearance provisions and foreclosure moratoria allowed distressed homeowners to defer payments and remain in their homes, reducing foreclosures and forced sales.

[21] Bank-year fixed effects absorb the exposure measure and its interaction with *Post*, while census tract-year fixed effects absorb *CRA* and *CRA* × *Post*.

Because foreclosures affect local house prices and supply of homes for sale, these policies could change mortgage applications and originations even if banks did not change credit supply (Gupta, 2019; Mian et al., 2015). This concern is particularly relevant in LMI census tracts because forbearance take-up was higher among lower-income borrowers and in communities with larger minority populations (Cherry et al., 2021). Banks that entered the pandemic with greater exposure to LMI census tracts may therefore have experienced larger forbearance-related changes in housing inventory and mortgage demand.

I proxy for this exposure using *High LMI Market Exposure*. I first calculate the share of each bank's 2018–2019 mortgage origination volume in LMI census tracts, *LMI Market Exposure*. I then define *High LMI Market Exposure* as an indicator equal to 1 when this share exceeds the median across banks, and zero otherwise. The indicator identifies banks that entered the pandemic with greater exposure to mortgage markets in which forbearance and foreclosure protections were more likely to affect housing activity.

A second concern is that the Federal Reserve's large-scale purchases of agency MBS affected mortgage supply differently across banks. Prior research finds that banks with larger MBS holdings increased mortgage lending more following Federal Reserve purchases of agency MBS (Chakraborty et al., 2020; Rodnyansky and Darmouni, 2017). Consistent with this literature, I measure a bank's MBS exposure as the sum of held-to-maturity MBS at amortized cost and available-for-sale MBS at fair value, divided by total assets and averaged over 2018–2019. *High MBS Holdings* equals 1 when the bank's pre-pandemic average exceeds the median across banks, and zero otherwise.

A related concern is that the pandemic-era support for the agency mortgage market affected banks differently according to their reliance on GSE securitization. Banks that routinely sold mortgages to Fannie Mae or Freddie Mac could finance new originations through the agency market rather than retain the loans on their balance sheets. Access to securitization reduces the balance-sheet capacity required

for mortgage lending (Fuster and Vickery, 2015; Loutskina and Strahan, 2009) and may also affect screening incentives because the originating lender does not retain the full credit risk of a sold loan (Keys et al., 2012).

I measure this exposure using *High GSE Securitization Exposure*. I first calculate the share of the bank's 2018–2019 mortgage origination volume sold to Fannie Mae or Freddie Mac, defined as the amount of such sales divided by the bank's total mortgage origination amount over the same period. *High GSE Securitization Exposure* equals 1 when this share exceeds the median across banks, and zero otherwise. The indicator identifies banks that entered the pandemic with greater reliance on agency securitization and the OTD model. Additionally, because mortgages purchased by Fannie Mae and Freddie Mac were eligible for the CARES Act forbearance, the indicator may also capture a bank's exposure to shocks related to forbearance.

Table 5 reports the results.[22] Panel A examines mortgage approvals, and Panel B examines interest rates on originated mortgages. Columns (1) through (3) allow post-2020 change in mortgage lending to LMI census tracts to vary with, respectively, LMI market exposure, MBS holdings, and GSE securitization exposure.[23] Column (4) includes all three exposure measures jointly. In Panel A, the coefficient on $CECL \times CRA \times Post$ ranges from -0.010 to -0.014 and remains statistically significant at the 5 percent level. In Panel B, the coefficient ranges from 0.029 to 0.030 percentage points, and remains statistically significant at the 5 percent level. The estimates are similar to the baseline results, indicating that the effects of differential exposure to these policies do not explain my findings. Figure 3 plots the dynamic effects controlling for three exposure measures jointly and shows no evidence of pre-trend. The

[22] Appendix B reports summary statistics for the continuous exposure measures. The results are similar when the exposure measures are specified continuously.
[23] The numbers of observations are lower in Columns (1) and (3) because some banks did not originate mortgages during the pre-period and have missing values for the corresponding exposure measures.

effect on approval rates appears in 2020 and remains negative thereafter, while the effect on interest rates emerges in 2021 and remains positive in 2022.

[Insert Table 5 near here]

[Insert Figure 3 near here]

5.2.2 Placebo Tests

I conduct three placebo analyses to assess whether the findings reflect severe economic stress, contemporaneous policy interventions, or differences between CECL and non-CECL banks rather than CECL adoption.

First, I examine mortgage lending during the 2007/2009 great financial crisis (GFC). Like the COVID-19 recession, the GFC was a severe downturn accompanied by fiscal and monetary interventions, but it did not coincide with CECL implementation. If banks that later adopted CECL responded differently to severe economic downturns or associated policy interventions, they might also have reduced lending in LMI relative to non-LMI tracts during the GFC. I re-estimate the dynamic specification from 2007 to 2009, using 2007 as the base year.[24] Figure 4 plots the coefficient estimates and their 90 percent confidence interval. The coefficients are statistically indistinguishable from zero during the GFC recession period, which started in December 2007 and ended in the third quarter of 2009. Thus, banks that later adopted CECL did not exhibit a decline in mortgage approvals in LMI tracts during the GFC.

[24] I use 2007 as the first year because the loan-level files obtained from CFPB's historical HMDA download portal begin in 2007.

[Insert Figure 4 near here]

Second, I examine federally insured or guaranteed mortgages. Federal insurance and guarantees absorb a substantial portion of credit losses on Federal Housing Administration (FHA) and Farm Service Agency/Rural Housing Service (FSA/RHS) loans, weakening the channel through which CECL affects mortgage supply.[25] I re-estimate the entropy-balanced model specified in Eq. (1) using these mortgages. Panel A of Table OA.3 in Online Appendix OA.3 reports economically small and statistically insignificant effects on approvals and interest rates.

Third, I examine lending under the Small Business Administration (SBA) 7(a) program, in which the SBA guarantees up to 85 percent of the loan amount for loans of $150,000 or less and up to 75 percent for loans above $150,000. The SBA data, however, report the borrower's ZIP code rather than its census tract. I construct a continuous measure of each ZIP code's exposure to LMI census tracts and aggregate SBA lending to the bank–ZIP code–quarter level and define the dependent variable as the natural log of total SBA loan amount originated by a bank in a ZIP code during the quarter.[26] I estimate an entropy-balance-weighted triple-difference specification, replacing *CRA* with the continuous ZIP-level LMI share. Panel B of Table OA.3 shows no evidence that CECL banks reduced SBA lending disproportionately in ZIP codes with greater exposure to LMI census tracts after CECL adoption.[27]

[25] FHA insurance covers 100% of the outstanding loan balance, the USDA Rural Housing Service's Section 502 program provides lenders with a 90% loan-note guarantee, and Farm Service Agency guarantees may cover up to 95% of losses on principal and interest.
[26] The details appear in Online Appendix OA.3.
[27] This test should be interpreted with caution. ZIP codes often cover multiple census tracts, so the weighted LMI share measures the composition of the ZIP code rather than the LMI status of the tract in which a borrower is located. The SBA analysis is less precise than the tract-level HMDA analysis. I interpret the null SBA result as a lower-powered placebo that provides corroborating evidence.

5.2.3 Corroborating Evidence from the 2023 CECL Adoption Banks

As a corroborating analysis, I examine banks that adopted CECL in 2023. This setting shifts the treatment date away from the onset of the pandemic while retaining a comparison among banks operating under the CRA requirements.

For this analysis, I use HMDA applications submitted to banks from 2021 to 2024. The treatment group consists of banks that adopted CECL in 2023, including private banks and public banks qualifying as SRCs, while the control group consists of large public banks that had adopted CECL in 2020. I define $Post_1$ as an indicator equal to 1 in 2023 and 2024, and zero otherwise. I restrict the sample to banks with total assets between $5 billion and $700 billion in 2022, excluding very small banks and the largest, most complex banks. This restriction retains a sufficient number of control banks for entropy balancing and avoids using extreme balancing weights. I entropy balance the 2020 adopters to the 2023 adopters on the same set of bank characteristics, and then estimate the specification as in Eq. (1). Table OA.4 in Online Appendix OA.4 reports the entropy balancing statistics.

Table 6 reports the results. Following CECL adoption in 2023, the later-adopting banks exhibited a 1.2 percent decline in approval rates in LMI census tracts relative to the differential changes at banks that adopted CECL in 2020. Interest rates on originated mortgages increased by 13.5 basis points in these areas. Figure OA.4 plots the dynamic effects, showing no differential pre-trend, with the effect appearing beginning in 2023.

[Insert Table 6 near here]

These results provide corroborating evidence that the main results are associated with CECL adoption rather than solely with the pandemic. The estimates should nevertheless be interpreted

cautiously and considered as supporting evidence because the control banks had already adopted CECL in 2020 and their lending response may have continued to evolve during the analysis period.

5.2.4 Geographic Comparability

Another concern is that CECL and non-CECL banks may operate in different geographic areas and face different local shocks. The concern is limited in the main sample: 93 percent of counties and 85 percent of census tracts have mortgage applications submitted to both CECL and non-CECL banks. To further strengthen geographic comparability, I re-estimate Eq. (1) after restricting the sample to counties in which both CECL and non-CECL banks are active (Table OA.5, Panel A) and to census tracts in which both types of banks are active (Table OA.5, Panel B). The estimated effects remain similar under both restrictions.

5.2.5 Alternative Size Threshold and External Validity

The $30 billion asset threshold improves comparability and common support between CECL and non-CECL banks. As a robustness test, I raise the cutoff to $50 billion, thereby including banks through the mid-sized range. The estimates and dynamic patterns, reported in Online Appendix OA.6, remain similar to the main results. This evidence supports the applicability of the results to the small and mid-sized banks.

The economic mechanism of the results may also apply to larger banks. CECL requires recognition of expected lifetime credit losses, and mortgages with greater expected default risk generate larger allowances and capital costs at origination. As larger banks are also subject to CRA requirements, they face the same trade-off between the capital cost of lending in LMI communities and the regulatory costs of reducing such lending. However, the magnitude of the estimated effect might not be extrapolated to larger banks. Larger banks have more diversified portfolios and funding sources, greater access to

securitization, and different capital and regulatory environments, all of which may affect how strongly CECL changes mortgage lending in LMI census tracts.

5.2.6 Non-CECL Banks' Mortgage Lending in LMI Census Tracts

A remaining concern is that non-CECL banks increased the supply of mortgage credit in LMI census tracts, and if so, the magnitude of coefficient estimates is biased upward. CECL banks reduce their mortgage credit supply in LMI census tracts following CECL adoption, but non-CECL banks might expand mortgage lending in these areas if such loans represent profitable investment opportunities. However, in anticipation of their own CECL adoption at the end of 2022, non-CECL banks might have reduced mortgage credit supply in LMI census tracts. Overall, I expect that non-CECL banks did not increase their supply of mortgage credit in LMI census tracts.

To examine whether non-CECL banks increased their supply of mortgage credit in LMI census tracts, I employ a DiD estimator comparing approval rates in LMI and non-LMI census tracts before and after 2020. Column (1) of Table OA.7 in Online Appendix OA.7 reports results, suggesting no significant change. To address the concern that divergent economic trends between LMI and non-LMI census tracts biased the coefficient estimates, I restrict the sample to mortgage applications from non-CECL banks in census tracts with an MFI between 70 and 90 percent of the surrounding area's MFI. Column (2) of Table OA.7 reports results, with no significant change apparent.

6. Non-bank Lenders' Mortgage Lending Response

6.1 Empirical Design and Results

I next test H2A by examining whether non-bank lenders increased the quantity of mortgage credit in LMI census tracts with greater exposure to CECL banks after CECL adoption. I then test H2B by

examining changes in interest rates for mortgage loans originated by non-bank lenders in these areas. Given findings reported in Section 5.1 that CECL banks reduced mortgage credit in LMI census tracts, these analyses focus on how non-bank lenders adjusted lending in these areas based on varying degrees of exposure to CECL banks.

I measure a census tract's exposure to CECL banks using *CECL Share*, the amount of mortgage loans originated by CECL banks divided by the total amount of mortgage loans originated by CECL banks, non-CECL banks, and non-bank lenders in a census tract from 2018 to 2019 (i.e., the pre-period). To ensure statistically meaningful calculations, I exclude census tracts with fewer than 30 mortgage loans originated in the pre-period.[28] Figure 5 shows examples of the geographic variation of census tracts' exposure to CECL banks. Panel A contains a map of census tracts' exposure to CECL banks for Orange County in California, and Panel B contains a map of census tracts' exposure to CECL banks for Fulton County in Georgia. Census tracts with red boundaries are LMI census tracts.

[Insert Figure 5 near here]

To test H2A, I restrict the sample to applications submitted to non-bank lenders and estimate the following triple-interaction estimator with a continuous exposure measure that examines whether, following CECL adoption, non-bank lenders increase mortgage approval rates more in LMI census tracts with greater exposure to CECL banks than in non-LMI census tracts:

$$\begin{aligned} I(Approved)_{i,l,c,s,t} = \; & \alpha + \beta_1 CECL\ Share_c \times CRA_c \times Post_t \\ & + \beta_2 CECL\ Share_c \times Post_t + \beta_3 CRA_c \times Post_t \\ & + \Gamma' X_{i,t} + \delta_{l,t} + \mu_{s,t} + \eta_c + \xi_{l,s} + \epsilon_{i,l,c,s,t}. \end{aligned} \quad (2)$$

[28] For statistical inference, a minimum of 30 observations is required to ensure that the sample mean distribution approximates normality, as the central limit theorem suggests. Results remain robust when all census tracts are included in the sample.

where i, l, c, s, and t index mortgage applications, lenders, census tracts, counties, and years, respectively. The dependent variable, $I(Approved)_{i,l,c,s,t}$ equals 1 if lender l approved loan application i in census tract c during year t, and zero otherwise. CECL Share$_c$ is the share of mortgage loans originated by CECL banks in census tract c during the pre-period. CRA$_c$ equals 1 for an LMI census tract, and zero otherwise. $Post_t$ equals 1 for years beginning from 2020, and zero for prior years.

Vector $X_{i,t}$ includes borrower and loan characteristics identical to those controlled in Eq. (1). I include lender-year and county-year fixed effects to control for time-varying lender characteristics and economic conditions in a county that might affect mortgage approval. Census-tract fixed effects control for time-invariant differences across tracts, while lender–county fixed effects control for differences in each lender's activities across local markets. I cluster standard errors at the lender level.

The lower-order interactions isolate the coefficient of interest from post-period changes associated with a census tract's exposure to CECL banks and its LMI status. The interaction *CECL Share* × *Post* absorbs post-2020 changes in non-bank approval rates that vary with a tract's pre-period exposure to CECL banks. For example, census tracts more exposed to CECL banks may have relied more heavily on bank lending or in-person mortgage origination before the pandemic and may have experienced a larger expansion in non-bank or internet-based lending as origination became more remote. *CRA* × *Post* absorbs post-period changes that are common across LMI census tracts.

Columns (1) and (2) of Table 7 report results. Both columns include lender-year, county-year, and census tract fixed effects. Column (2) adds lender-county fixed effects. The coefficient on *CECL Share* × $CRA \times Post$ is positive and statistically significant in both columns. The estimate in Column (2) suggests that following CECL adoption, increasing *CECL Share* from the first quartile (0.193) to the third quartile (0.365) is associated with a 0.3 percent larger increase in approval rates in LMI census tracts than in non-LMI census tracts. Loutskina and Strahan (2009) document that moving

bank balance sheet liquidity from the first to the third quartile increases approval rates by 0.6 percent. The economic magnitude of the observed effect is half that magnitude and economically meaningful. In the sample, the total volume of mortgage applications in LMI census tracts submitted to non-bank lenders in 2019 was 85.43 billion. A 0.3 percent increase in mortgage approval rates translated to an economically significant 256 million increase in mortgage credit supply from non-bank lenders to households in LMI census tracts exposed more to CECL banks.

To test whether non-bank lenders changed mortgage interest rates (H2B), I replace the dependent variable in Eq. (2) with *Interest Rate*, focusing on originated loans and including the same set of controls and fixed effects as in Columns (1) to (2) of Table 7. Columns (3) and (4) of Table 7 report results. Both columns suggest that CECL exposure had no effect on non-bank lenders' mortgage interest rates.

[Insert Table 7 near here]

To assess pre-trends, I estimate a modified version of Eq. (2), replacing $Post$ with year indicators for each year in the triple interaction term *CECL Share* $\times$ $CRA \times Post$, where the dependent variables are *I(Approved)* and *Interest Rate*. Panel A of Figure 6 plots coefficient estimates for approval rates, which do not appear to trend differently pre-CECL. Coefficients for the post period are positive and statistically significant, consistent with non-bank lenders expanding mortgage approvals in LMI tracts with greater pre-period exposure to CECL banks. Panel B of Figure 6 plots coefficient estimates for interest rates, suggesting no significant change in risk-adjusted interest rates of mortgages originated by non-bank lenders in LMI census tracts more exposed to CECL banks during 2020 and 2021. However, the coefficient for 2022 is positive and statistically significant. One interpretation is that non-bank lenders initially accommodated the lending gap created by CECL banks' withdrawal without materially changing risk-adjusted loan pricing. By 2022, however, the cumulative reallocation of LMI borrowers

toward non-bank lenders might have increased the cost or credit risk of serving these borrowers, leading non-bank lenders to charge higher risk-adjusted interest rates.

[Insert Figure 6 near here]

Results reported in this section suggest that non-bank lenders increased mortgage credit supply to LMI census tracts that were exposed more to CECL banks after CECL adoption. There is little evidence that non-bank lenders increase interest rates in LMI census tracts that were exposed more to CECL banks.

6.2 Additional Tests

As a placebo test, I re-estimate Eq. (2) using FHA and FSA/RHS mortgages. The results, reported in Online Appendix OA.8 Table OA.8, show that the estimated coefficients on *CECL Share* × *CRA* × *Post* are economically small and statistically insignificant for both approval rates and interest rates.

I also test whether non-CECL banks increased mortgage credit supply in LMI census tracts more exposed to CECL banks. I replace the main test variable with *CECL Share* × $Post$ and estimate the model within mortgage applications in LMI census tracts. Column (3) of Table OA.7 in Online Appendix OA.7 reports results, suggesting non-CECL banks did not increase mortgage credit supply in LMI census tracts that were exposed more to CECL banks.

7. Conclusion

This paper assesses how adoption of the CECL accounting standard impacted geographic disparities in mortgage credit access. I find that CECL adoption had important distributive consequences on credit access across LMI and non-LMI census tracts. To identify the effects of CECL adoption on

mortgage supply in LMI census tracts across CECL banks, I implement a triple-difference strategy that allows comparisons between CECL and non-CECL banks in LMI and non-LMI census tracts. I entropy balance on several bank characteristics. As hypothesized, CECL adoption led to declines in mortgage credit supply by adopting banks in LMI census tracts.

I examine how non-bank lenders responded to the contraction of mortgage credit by CECL banks in LMI census tracts. I use a triple-interaction estimator with a continuous exposure measure that compares changes in mortgage approval rates of non-bank lenders in LMI versus non-LMI census tracts before and after CECL adoption and tests whether this differential change is greater in census tracts with higher exposure to CECL banks. I find that non-bank lenders mitigated reductions to mortgage credit in LMI census tracts that were more exposed to CECL adoption.

This paper evidences how a change in banks' accounting regimes interacted with CRA to influence geographic inequality in financial inclusion. I highlight that CECL adoption led to social costs by reducing banks' promotion of equitable credit access under CRA. I also show that non-bank lenders mitigated reductions in credit supply by banks in these underserved areas by lending to borrowers served previously by CECL banks with higher credit quality. Results suggest that extending CRA lending requirements to non-bank lenders may be necessary to promote credit supply to LMI households that experience reduced access due to regulatory changes that affect banks.

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A. Variable Definitions

Table A.1: Variable Definitions

Variable	Definition
Variables of Interest	
CECL	An indicator that takes a value of 1 if the bank has adopted CECL in 2020, and zero otherwise.
CRA	An indicator that takes a value of 1 if the census tract was an LMI tract (i.e., tract with median family income less than 80 percent of the MSA/MD median family income or the state non-metropolitan area median family income for tracts located in non-metropolitan areas), and zero otherwise.
Post	An indicator that takes a value of 1 if beginning in 2020, and zero for prior years.
Post_1	An indicator that takes a value of 1 if beginning in 2023, and zero for prior years.
Treat	An indicator that takes a value of 1 for banks that adopted CECL in 2023 and 0 for banks that adopted CECL in 2020.
CECL Share	The amount of mortgage loans originated by CECL banks divided by the amount of mortgage loans originated by all CECL, non-CECL, and non-bank lenders in a census tract from 2018 to 2019. I exclude census tracts with fewer than 30 mortgage loans originated.
Loan-Level Variables	
I(Approved)	An indicator that takes a value of 1 if a loan was approved, and zero otherwise.
Log(Income)	The natural logarithm of the applicant's income.
Log(Loan Amount)	The natural logarithm of the loan amount.
LTV	The ratio of the loan amount and the property value in percent.
I(DTI>36%)	An indicator that equals 1 if an applicant's DTI ratio was larger than 36 percent, and zero otherwise.
Interest Rate	The interest rate of the loan in percent.
Conforming	An indicator that equals 1 if a loan amount was less than the conforming loan limit, and zero otherwise.
Owner Occupied	An indicator that equals 1 if a property was owner-occupied as a principal residence and zero if as a second residence or investment property.
Loan Purpose	An indicator that equals 1 if a loan was for home purchase and zero if it was for refinancing.
Female	An indicator that equals 1 if an applicant's gender was female, and zero otherwise.
White	An indicator that equals 1 if an applicant's race was white, and zero otherwise.
Bank-Year Variables	
Log(Size)	The natural logarithm of total assets.
Tier-1 Capital Ratio	Tier-1 capital divided by total assets.
Deposit Ratio	Total deposits divided by total assets.
Residential R/E Loan Ratio	Amount of residential real estate loans divided by total assets.

C&I Loan Ratio	Amount of commercial and industrial loans divided by total assets.
NPL Ratio	Amount of loans that are 90 days or more past due and still accruing and those in nonaccrual status divided by total assets.
Bank-Level Variables	
High LMI Market Exposure	An indicator that equals 1 if the bank's *LMI Market Exposure* exceeds the median across banks, and zero otherwise. *LMI Market Exposure* is calculated as the dollar amount of mortgages originated by a bank in LMI census tracts during 2018–2019 divided by the bank's total dollar amount of mortgage originations during the same period.
High MBS Holdings	An indicator that equals 1 if the bank's average *MBS Holdings Ratio* during 2018–2019 exceeds the median across banks, and zero otherwise. *MBS Holdings Ratio* is calculated as the sum of held-to-maturity mortgage-backed securities measured at amortized cost and available-for-sale mortgage-backed securities measured at fair value, divided by total assets and averaged over 2018–2019.
High GSE Securitization Exposure	An indicator that equals 1 if the bank's *GSE Securitization Exposure* exceeds the median across banks, and zero otherwise. *GSE Securitization Exposure* is calculated as the dollar amount of mortgages originated by a bank during 2018–2019 and sold to Fannie Mae or Freddie Mac divided by the bank's total dollar amount of mortgage originations during the same period.

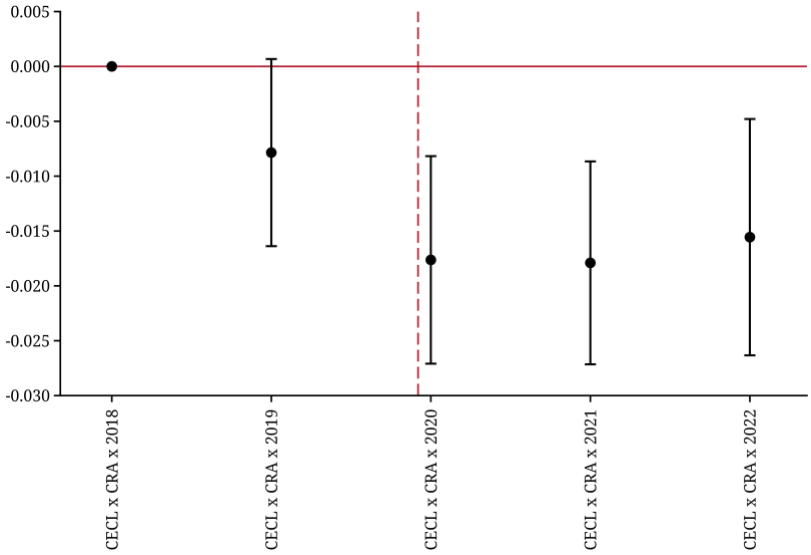
B. Summary Statistics

Table B.1: Summary Statistics for Bank-Level Heterogeneity Measures

Variable	N	Mean	SD	P25	Median	P75
LMI Market Exposure	1,091	0.104	0.088	0.054	0.083	0.083
GSE Securitization Exposure	1,091	0.129	0.224	0.000	0.000	0.000
MBS Holdings Ratio	1,331	0.072	0.072	0.017	0.057	0.105

This table reports bank-level summary statistics for the continuous exposure measures. *LMI Market Exposure* is calculated as the dollar amount of mortgages originated by a bank in LMI census tracts during 2018–2019 divided by the bank's total dollar amount of mortgage originations during the same period. *GSE Securitization Exposure* is calculated as the dollar amount of mortgages originated by a bank during 2018–2019 and sold to Fannie Mae or Freddie Mac divided by the bank's total dollar amount of mortgage originations during the same period. *MBS Holdings Ratio* is calculated as the sum of held-to-maturity mortgage-backed securities measured at amortized cost and available-for-sale mortgage-backed securities measured at fair value, divided by total assets and averaged over 2018–2019.

Panel A: I(Approved)



Panel B: Interest Rate

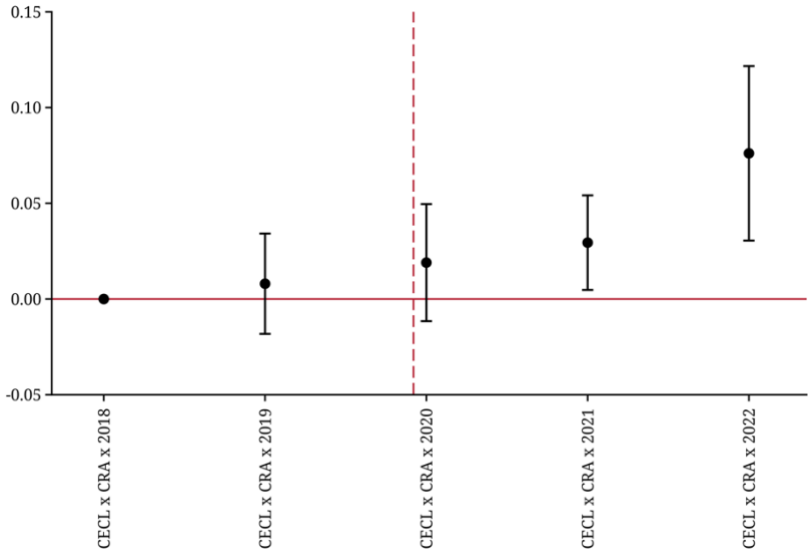


Fig. 1: Dynamic Effects of CECL Adoption on Bank Mortgage Credit Supply. This figure plots coefficient estimates on year-by-year indicators and their 90% confidence intervals from dynamic entropy-balanced-weighted triple-difference estimations of Eq. (1), in which dependent variables are *I(Approved)* and *Interest Rate*. The sample includes mortgage applications submitted to banks in Panel A and mortgages originated by banks in Panel B. Coefficient estimates are reported relative to 2018. The regression includes bank-year, census tract-year, CECL bank-census tract, and bank-county fixed effects, and all borrower and loan characteristics control variables. Standard errors are clustered by banks. All variables are defined in Appendix A.

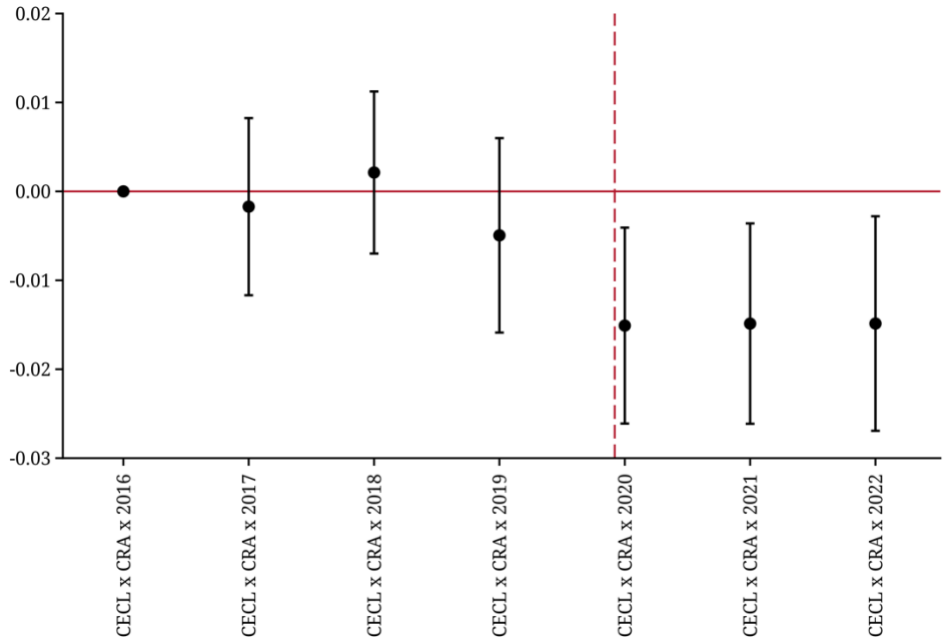
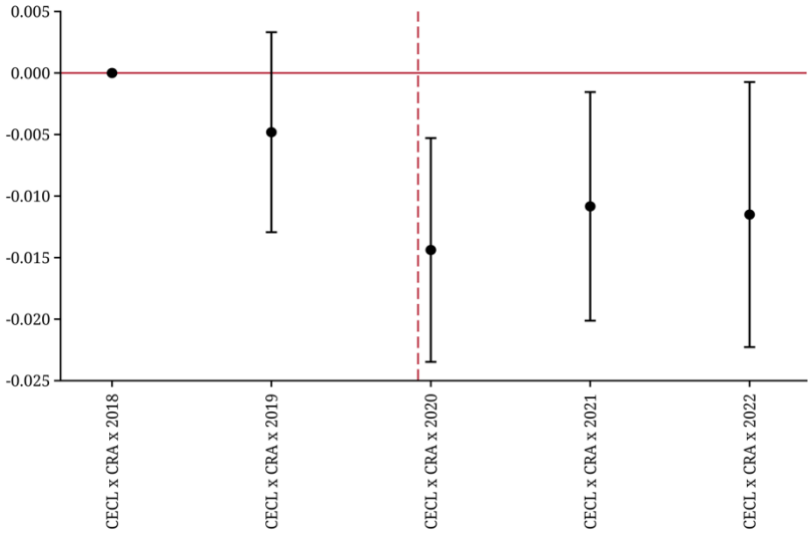


Fig. 2: Dynamic Effects of CECL Adoption on Bank Mortgage Approval Extended. This figure plots coefficient estimates on year-by-year indicators and their 90% confidence intervals from dynamic entropy-balanced-weighted triple-difference estimations of Eq. (1), in which dependent variable is *I(Approved)*. The sample extends to mortgage applications submitted to banks from 2016 to 2022. Coefficient estimates are reported relative to 2016. The regression includes all borrower and loan characteristics control variables and fixed effects used in Eq. (1), except for *LTV* and *I(DTI>36%)*. Standard errors are clustered by banks. All variables are defined in Appendix A.

Panel A: I(Approved)



Panel B: Interest Rate

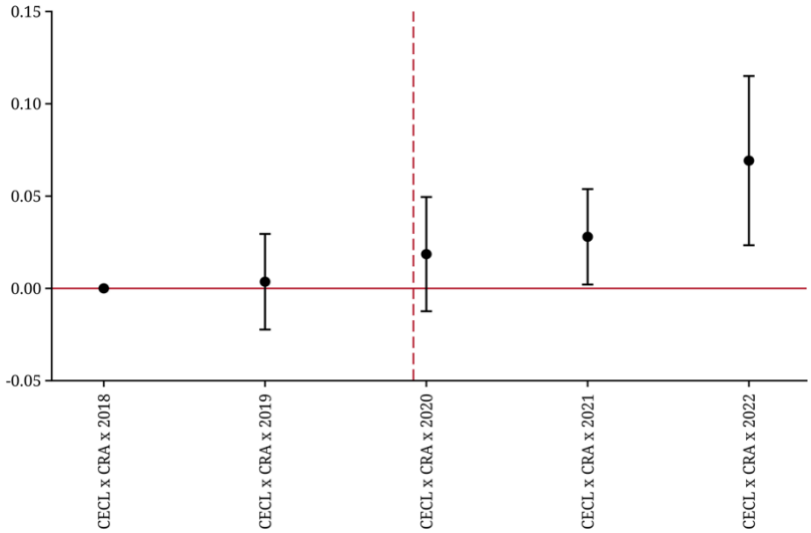


Fig. 3: Dynamic Effects of CECL Adoption after Controlling for Exposure to Pandemic Policies. This figure plots coefficient estimates on year-by-year indicators and their 90% confidence intervals from dynamic entropy-balance-weighted triple-difference estimations of Eq. (1). The regressions control for *High LMI Market Exposure*, *High MBS Holdings*, and *High GSE Securitization Exposure*, each interacted with *CRA* and *CRA* × *Post*. The dependent variables are *I(Approved)* in Panel A and *Interest Rate* in Panel B. The sample includes mortgage applications submitted to banks in Panel A and mortgages originated by banks in Panel B. Coefficient estimates are reported relative to 2018. The regressions also include bank-year, census tract-year, CECL bank-census tract, and bank-county fixed effects, as well as all borrower and loan characteristics control variables. Standard errors are clustered by banks. All variables are defined in Appendix A.

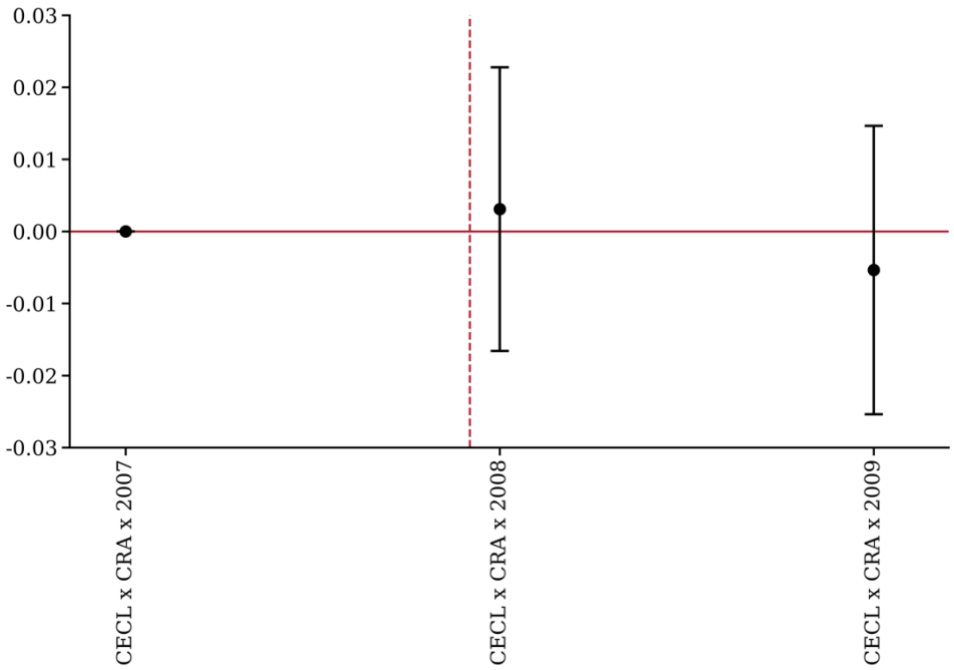
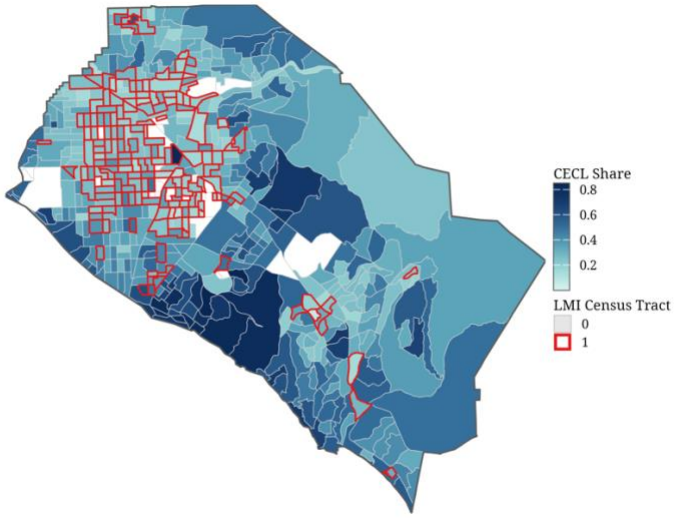


Fig. 4: Placebo Test Using the 2007/2009 Financial Crisis. This figure plots coefficient estimates on year-by-year indicators and their 90% confidence intervals from dynamic triple-difference estimations of Eq. (1) using the 2007/2009 financial crisis as a placebo event. The dependent variable is *I(Approved)*. The sample includes mortgage applications submitted to banks during 2007–2009. Coefficient estimates are reported relative to 2007. The regressions include bank-year, census tract-year, CECL bank-census tract, and bank-county fixed effects, as well as all borrower and loan characteristics control variables. Standard errors are clustered by banks. All variables are defined in Appendix A.

Panel A: Orange County



Panel B: Fulton County

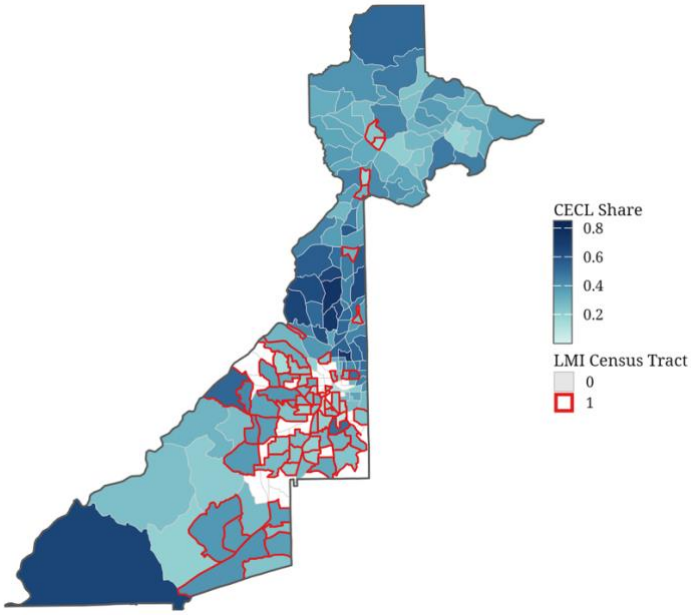
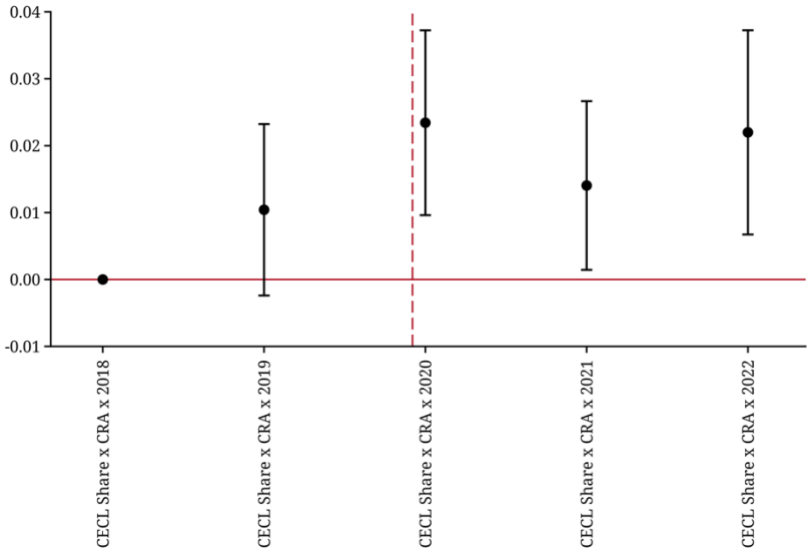


Fig. 5: Examples of Census Tracts' Exposure to CECL Banks. This figure plots maps for census tracts' exposure to CECL banks for Orange County in California and Fulton County in Georgia. Blue represents census tracts with different degrees of exposure to CECL banks, measured by *CECL Share*. Census tracts with red boundaries are LMI census tracts. I exclude census tracts with fewer than 30 mortgage loans originated in the pre-period for a statistically meaningful calculation of *CECL Share* for a census tract. Census tracts with fewer than 30 mortgage loans that originated during the pre-period are in white. All variables are defined in Appendix A.

Panel A: I(Approved)



Panel B: Interest Rate

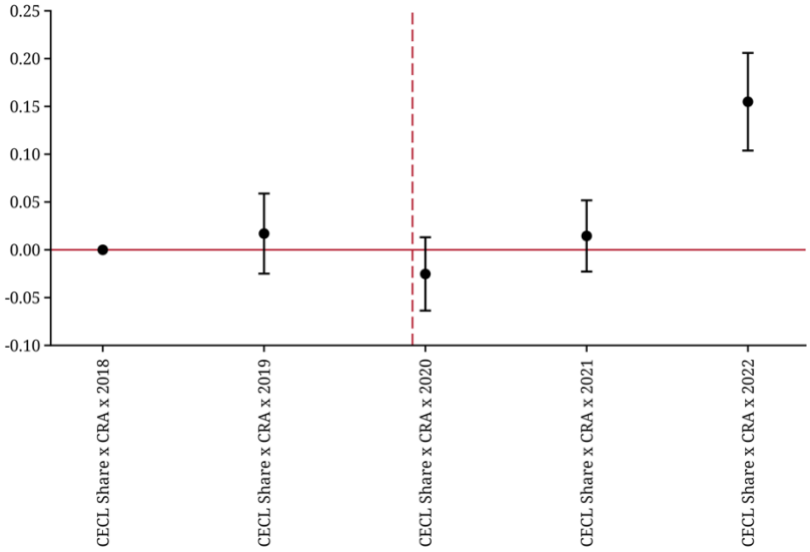


Fig. 6: Dynamic Effects of CECL Exposure on Non-bank Lender Mortgage Credit Supply. This figure plots coefficient estimates on year-by-year indicators and their 90% confidence intervals from dynamic estimations of Eq. (2), in which dependent variables are *I(Approved)* and *Interest Rate*. The sample includes mortgage applications submitted to (Panel A) and originated by (Panel B) non-bank lenders in LMI census tracts. Coefficient estimates are reported relative to 2018. The regression includes lender-year, county-year, census tract, and lender-county fixed effects, as well as all borrower and loan characteristics control variables. Standard errors are clustered by lenders. All variables are defined in Appendix A.

Table 1: Sample Selection

	# of Loan Applications	# of Lenders	# of Census Tracts
Panel A: Sample selection for banks			
1) HMDA conventional loan applications with an amount larger than $1,000 from banks, either approved or denied, and supervised by OCC, FRS, FDIC, or CFPB from 2018 to 2022	16,636,776	3,094	72,786
2) Require valid values for loan characteristics, including gender, race, income, loan conforming status, occupancy type, loan purpose, and census tract	13,596,046	3,065	72,567
3) Require valid values for loan characteristics, including LTV and DTI ratios	12,288,009	1,476	72,524
4) Require valid values for bank characteristics	12,251,761	1,432	72,517
5) Exclude banks that adopted CECL between 2021 and 2022	11,842,641	1,378	72,512
6) Exclude CECL banks with total assets greater than $30 billion	5,402,941	1,331	71,582
Panel B: Sample selection for non-bank lenders			
1) HMDA conventional loan applications with amount larger than $1,000 from non-bank lenders, as defined in Avery et al. (2007), from 2018 to 2022	24,892,993	1,154	72,906
2) Require valid values for loan characteristics, including gender, race, income, loan conforming status, occupancy type, loan purpose, and census tract	19,261,586	1,067	72,712
3) Require valid values for loan characteristics, including LTV and DTI ratios	18,114,341	1,024	72,096
4) Require valid values for *CECL Share*	17,528,179	1,021	55,607

This table reports the sample selection procedure. Panel A details selection for mortgage application records at banks, and Panel B for that at non-bank lenders.

Table 2: Summary Statistics of Mortgage Applications

Panel A: Mortgage Applications from Banks

	Non-LMI Census Tracts				LMI Census Tracts				All Census Tracts			
	Mean	Median	SD	N	Mean	Median	SD	N	Mean	Median	SD	N
I(Approved)	0.908	1.000	0.289	4,698,092	0.871	1.000	0.336	704,849	0.903	1.000	0.296	5,402,941
Log(Income)	11.584	11.552	0.706	4,698,092	11.276	11.225	0.688	704,849	11.544	11.513	0.712	5,402,941
Log(Loan Amount)	12.266	12.324	0.777	4,698,092	11.900	11.951	0.815	704,849	12.218	12.278	0.791	5,402,941
LTV	71.093	75.000	18.923	4,698,092	74.688	79.060	19.029	704,849	71.562	75.000	18.975	5,402,941
I(DTI>36%)	0.415	0.000	0.493	4,698,092	0.464	0.000	0.499	704,849	0.421	0.000	0.494	5,402,941
White	0.883	1.000	0.321	4,698,092	0.801	1.000	0.399	704,849	0.873	1.000	0.333	5,402,941
Female	0.194	0.000	0.396	4,698,092	0.274	0.000	0.446	704,849	0.205	0.000	0.403	5,402,941
Conforming	0.936	1.000	0.245	4,698,092	0.984	1.000	0.127	704,849	0.942	1.000	0.234	5,402,941
Owner Occupied	0.898	1.000	0.302	4,698,092	0.814	1.000	0.389	704,849	0.887	1.000	0.316	5,402,941
Loan Purpose	0.466	0.000	0.499	4,698,092	0.551	1.000	0.497	704,849	0.477	0.000	0.499	5,402,941
Interest Rate	3.743	3.500	1.063	4,258,056	4.016	3.790	1.173	611,228	3.777	3.500	1.081	4,869,284
CECL									0.368	0.000	0.482	5,402,941
CRA									0.130	0.000	0.337	5,402,941
Post									0.670	1.000	0.470	5,402,941

Table 2 continued from previous page

Panel B: Mortgage Applications from Non-bank Lenders

	Non-LMI Census Tracts				LMI Census Tracts				All Census Tracts			
	Mean	Median	SD	N	Mean	Median	SD	N	Mean	Median	SD	N
I(Approved)	0.908	1.000	0.289	14,970,240	0.885	1.000	0.318	2,557,939	0.905	1.000	0.294	17,528,179
Log(Income)	11.559	11.552	0.601	14,970,240	11.315	11.290	0.600	2,557,939	11.524	11.513	0.607	17,528,179
Log(Loan Amount)	12.482	12.525	0.619	14,970,240	12.221	12.278	0.731	2,557,939	12.444	12.487	0.643	17,528,179
LTV	72.156	75.000	17.789	14,970,240	74.688	76.970	18.012	2,557,939	72.525	75.000	17.844	17,528,179
I(DTI>36%)	0.483	0.000	0.500	14,970,240	0.543	1.000	0.498	2,557,939	0.492	0.000	0.500	17,528,179
White	0.817	1.000	0.387	14,970,240	0.769	1.000	0.421	2,557,939	0.810	1.000	0.392	17,528,179
Female	0.230	0.000	0.421	14,970,240	0.293	0.000	0.455	2,557,939	0.239	0.000	0.427	17,528,179
Conforming	0.972	1.000	0.165	14,970,240	0.994	1.000	0.079	2,557,939	0.975	1.000	0.155	17,528,179
Owner Occupied	0.909	1.000	0.288	14,970,240	0.850	1.000	0.358	2,557,939	0.900	1.000	0.300	17,528,179
Loan Purpose	0.428	0.000	0.495	14,970,240	0.464	0.000	0.499	2,557,939	0.433	0.000	0.495	17,528,179
Interest Rate	3.611	3.375	1.006	13,537,218	3.785	3.500	1.071	2,242,784	3.636	3.375	1.018	15,780,002
CECL Share	0.301	0.274	0.141	14,970,240	0.249	0.228	0.116	2,557,939	0.294	0.267	0.139	17,528,179

This table reports summary statistics for loan and borrower characteristics of mortgage applications, by LMI census tracts, non-LMI census tracts, and all census tracts. Panel A reports descriptive statistics for loan and borrower characteristics of mortgage applications submitted to banks, and Panel B reports corresponding descriptive statistics for mortgage applications submitted to non-bank lenders. Continuous variables are winsorized at the 1% level. All variables are defined in Appendix A.

Table 3: Summary Statistics of Bank Characteristics

Panel A: Summary Statistics of Bank Characteristics

	N	Mean	SD	*p25*	Median	*p75*
Log(Size)	4,140	20.868	1.337	19.953	20.791	21.632
Deposit Ratio	4,140	0.829	0.061	0.797	0.843	0.873
Tier-1 Capital Ratio	4,140	0.103	0.024	0.088	0.097	0.110
Residential R/E Loan Ratio	4,140	0.234	0.154	0.126	0.193	0.296
C&I Loan Ratio	4,140	0.091	0.066	0.045	0.079	0.123
NPL Ratio	4,140	0.005	0.005	0.002	0.004	0.007

Panel B: Entropy Balancing Statistics of Bank Characteristics

	Treatment	Control	
	Mean	Pre-balancing Mean	Post-balancing Mean
Log(Size)	22.717	20.559***	22.717
Deposit Ratio	0.806	0.832***	0.806
Tier-1 Capital Ratio	0.096	0.104***	0.096
Residential R/E Loan Ratio	0.181	0.242***	0.181
C&I Loan Ratio	0.119	0.087***	0.119
NPL Ratio	0.005	0.005	0.005

This table reports summary and entropy balancing statistics of bank characteristics. CECL and non-CECL banks are balanced by means on characteristics, including *Log(Size)*, *Tier-1 Capital Ratio*, *Deposit Ratio*, *Residential R/E Loan Ratio*, *C&I Loan Ratio*, and *NPL Ratio*. Panel A reports summary statistics for banks, and Panel B reports entropy balancing statistics for banks, including statistics for non-CECL banks both before and after applying entropy balancing weights. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively, for tests of differences between the means of characteristics of CECL banks and non-CECL banks. Continuous variables are winsorized at the 1% level. All variables are defined in Appendix A.

Table 4: Effect of CECL Adoption on Bank Mortgage Credit Supply

	I(Approved)		Interest Rate	
	(1)	(2)	(3)	(4)
CECL × CRA × Post	-0.015***	-0.014***	0.032**	0.030**
	(0.004)	(0.004)	(0.014)	(0.013)
Log(Loan Amount)	0.030***	0.029***	-0.402***	-0.390***
	(0.007)	(0.008)	(0.067)	(0.070)
Log(Income)	0.069***	0.069***	0.097***	0.095***
	(0.008)	(0.008)	(0.024)	(0.025)
LTV	-0.002***	-0.002***	0.007***	0.007***
	(0.000)	(0.000)	(0.001)	(0.001)
I(DTI>36%)	-0.088***	-0.088***	0.134***	0.133***
	(0.008)	(0.008)	(0.016)	(0.016)
White	0.045***	0.044***	0.037***	0.036***
	(0.003)	(0.003)	(0.006)	(0.006)
Female	0.008***	0.007***	0.007**	0.007**
	(0.001)	(0.001)	(0.003)	(0.003)
Conforming	0.083***	0.083***	-0.121***	-0.113***
	(0.006)	(0.006)	(0.032)	(0.032)
Owner Occupied	0.057***	0.058***	-0.230***	-0.245***
	(0.008)	(0.009)	(0.031)	(0.032)
Loan Purpose	0.085***	0.083***	0.047**	0.042*
	(0.006)	(0.006)	(0.023)	(0.022)
Fixed Effects				
Bank-Year	Yes	Yes	Yes	Yes
Census Tract-Year	Yes	Yes	Yes	Yes
CECL Bank-Census Tract	Yes	Yes	Yes	Yes
Bank-County	No	Yes	No	Yes
Adj. R^2	0.273	0.271	0.620	0.626
Observations	5,402,941	5,402,941	4,735,885	4,735,885

This table reports entropy-balance-weighted triple-difference estimation results of Eq. (1). Dependent variables are *I(Approved)* in columns (1) and (2) and *Interest Rate* in columns (3) and (4). The sample consists of mortgage applications submitted to banks in columns (1) and (2) and mortgages originated by banks in columns (3) and (4). Standard errors appear in parentheses and are clustered by banks. ***, **, and * indicate statistical significance at 1%, 5%, and 10%, respectively. All variables are defined in Appendix A.

Table 5: Controlling for Exposure to Pandemic Policies

	(1)	(2)	(3)	(4)
Panel A: I(Approved)				
CECL × CRA × Post	-0.014***	-0.011***	-0.014***	-0.010**
	(0.004)	(0.004)	(0.004)	(0.004)
High LMI Market Exposure × CRA × Post	-0.001			-0.003
	(0.005)			(0.005)
High LMI Market Exposure × CRA	0.025***			0.018***
	(0.004)			(0.005)
High MBS Holdings × CRA × Post		-0.001		0.002
		(0.004)		(0.004)
High MBS Holdings × CRA		-0.016***		-0.019***
		(0.005)		(0.004)
High GSE Securitization Exposure × CRA × Post			-0.001	-0.007
			(0.004)	(0.007)
High GSE Securitization Exposure × CRA			0.016***	0.000
			(0.004)	(0.008)
Controls				
Loan and Borrower Characteristics	Yes	Yes	Yes	Yes
Fixed Effects				
Bank-Year	Yes	Yes	Yes	Yes
Census Tract-Year	Yes	Yes	Yes	Yes
CECL Bank-Census Tract	Yes	Yes	Yes	Yes
Bank-County	Yes	Yes	Yes	Yes
Adjusted R^2	0.270	0.271	0.270	0.270
Observations	5,200,377	5,402,941	5,200,377	5,200,377

Table 5 continued from previous page

	(1)	(2)	(3)	(4)
Panel B: Interest Rate				
CECL × CRA × Post	0.029**	0.029**	0.029**	0.030**
	(0.012)	(0.013)	(0.013)	(0.013)
High LMI Market Exposure × CRA × Post	0.005			-0.021
	(0.014)			(0.017)
High LMI Market Exposure × CRA	-0.019			0.014
	(0.013)			(0.014)
High MBS Holdings × CRA × Post		-0.042**		-0.042**
		(0.017)		(0.017)
High MBS Holdings × CRA		0.047***		0.047***
		(0.016)		(0.016)
High GSE Securitization Exposure × CRA × Post			0.004	0.086***
			(0.014)	(0.027)
High GSE Securitization Exposure × CRA			0.009	-0.090***
			(0.014)	(0.027)
Controls				
Loan and Borrower Characteristics	Yes	Yes	Yes	Yes
Fixed Effects				
Bank-Year	Yes	Yes	Yes	Yes
Census Tract-Year	Yes	Yes	Yes	Yes
CECL Bank-Census Tract	Yes	Yes	Yes	Yes
Bank-County	Yes	Yes	Yes	Yes
Adjusted R^2	0.625	0.626	0.625	0.625
Observations	4,551,945	4,735,885	4,551,945	4,551,945

This table reports the entropy-balance-weighted triple-difference estimations of Eq. (1) after controlling for *High LMI Market Exposure*, *High MBS Holdings*, and *High GSE Securitization Exposure*, each interacted with *CRA* and *CRA* × *Post*. Panel A reports results for mortgage approval, and Panel B reports results for the interest rate on originated mortgages. Column (1) controls for *High LMI Market Exposure*, Column (2) controls for *High MBS Holdings*, and Column (3) controls for *High GSE Securitization Exposure*. Column (4) includes all three exposure measures simultaneously. Standard errors, reported in parentheses, are clustered at the bank level. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. All variables are defined in Appendix A.

Table 6: Corroborating Analysis Using Banks Adopting CECL in 2023

	I(Approved)	Interest Rate
	(1)	(2)
Treat × CRA × $Post_1$	-0.012**	0.135***
	(0.005)	(0.041)
Log(Income)	0.071***	0.090***
	(0.006)	(0.027)
Log(Loan Amount)	0.015***	-0.169**
	(0.003)	(0.077)
LTV	-0.001***	0.003***
	(0.000)	(0.001)
I(DTI > 36%)	-0.069***	0.138***
	(0.007)	(0.019)
White	0.036***	0.066***
	(0.003)	(0.012)
Female	0.008***	0.006
	(0.002)	(0.005)
Conforming	0.063***	0.047
	(0.004)	(0.043)
Owner Occupied	0.057***	-0.289***
	(0.007)	(0.035)
Loan Purpose	0.075***	0.021
	(0.004)	(0.029)
Fixed Effects		
Bank-Year	Yes	Yes
Census Tract-Year	Yes	Yes
Treatment Bank-Census Tract	Yes	Yes
Bank-County	Yes	Yes
Adjusted R^2	0.253	0.787
Observations	3,387,924	2,762,929

This table presents a corroborating analysis using the 2023 adoption of CECL by smaller public and private banks. The treatment group consists of banks that adopted CECL in 2023, while the control group consists of banks that had adopted CECL in 2020. $Post_1$ equals 1 in 2023 and 2024, and zero otherwise. The dependent variable is *I(Approved)* in Column (1) and the interest rate on originated mortgages in Column (2). The sample in Column (1) consists of mortgage applications, whereas the sample in Column (2) consists of originated mortgages. Standard errors, reported in parentheses, are clustered by banks. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. All variables are defined in Appendix A.

Table 7: Effect of CECL Exposure on Non-bank Lender Mortgage Credit Supply

	I(Approved)		Interest Rate	
	(1)	(2)	(3)	(4)
CECL Share × CRA × Post	0.016***	0.015***	0.012	0.010
	(0.006)	(0.005)	(0.015)	(0.014)
CECL Share × Post	0.031***	0.030***	-0.006	-0.005
	(0.004)	(0.004)	(0.016)	(0.016)
CRA × Post	-0.007***	-0.006***	-0.022***	-0.021***
	(0.002)	(0.002)	(0.005)	(0.005)
Log(Income)	0.051***	0.051***	-0.051***	-0.050***
	(0.004)	(0.004)	(0.012)	(0.012)
Log(Loan Amount)	-0.002	-0.002	0.098***	0.096***
	(0.002)	(0.002)	(0.036)	(0.036)
LTV	-0.001***	-0.001***	0.004***	0.005***
	(0.000)	(0.000)	(0.000)	(0.000)
I(DTI>36%)	-0.034***	-0.033***	0.069***	0.067***
	(0.003)	(0.003)	(0.005)	(0.005)
White	0.026***	0.025***	0.055***	0.052***
	(0.002)	(0.002)	(0.004)	(0.005)
Female	0.004***	0.004***	0.028***	0.027***
	(0.001)	(0.000)	(0.002)	(0.002)
Conforming	0.119***	0.118***	-0.114***	-0.112***
	(0.009)	(0.009)	(0.033)	(0.033)
Owner Occupied	0.049***	0.049***	-0.420***	-0.424***
	(0.005)	(0.005)	(0.014)	(0.014)
Loan Purpose	0.058***	0.054***	0.108***	0.107***
	(0.007)	(0.006)	(0.009)	(0.008)
Fixed Effects				
Lender-Year	Yes	Yes	Yes	Yes
County-Year	Yes	Yes	Yes	Yes
Census Tract	Yes	Yes	Yes	Yes
Lender-County	No	Yes	No	Yes
Adj. R^2	0.142	0.154	0.639	0.648
Observations	17,528,179	17,528,179	15,305,554	15,305,554

This table reports triple-difference estimation results examining the effect of exposure to CECL banks on non-bank mortgage credit supply. Dependent variables are *I(Approved)* in columns (1) and (2) and *Interest Rate* in columns (3) and (4). The sample consists of mortgage applications submitted to non-bank lenders in columns (1) and (2) and mortgages originated by non-bank lenders in columns (3) and (4). Standard errors appear in parentheses and are clustered by lenders. ***, **, and * indicate statistical significance at 1%, 5%, and 10%, respectively. All variables are defined in Appendix A.

Online Appendix

OA.1. Allowance for Loan Losses on Residential R/E Loans

Table OA.1: Pre-period Evidence on Allowance for Loan Losses on Residential R/E Loans

Panel A: Summary Statistics						
	N	Mean	SD	p25	Median	p75
Residential R/E Allowance Ratio	3,095	0.002	0.002	0.001	0.001	0.002
LMI Market Exposure	3,095	0.095	0.052	0.064	0.087	0.116
Log(Total Assets)	3,095	21.939	0.988	21.184	21.652	22.514
Tier-1 Capital Ratio	3,095	0.101	0.020	0.089	0.097	0.107
Residential R/E Loan Ratio	3,095	0.213	0.137	0.121	0.186	0.261
C&I Loan Ratio	3,095	0.101	0.072	0.053	0.088	0.134
Deposit Ratio	3,095	0.806	0.063	0.770	0.815	0.852
NPL Ratio	3,095	0.006	0.010	0.002	0.004	0.006

Panel B: Regression Results	
	Residential R/E Allowance Ratio
	(1)
LMI Market Exposure	0.003**
	(0.001)
Lagged Log(Total Assets)	−0.000
	(0.000)
Lagged Tier-1 Capital Ratio	0.016
	(0.009)
Lagged Residential R/E Loan Ratio	0.005***
	(0.001)
Lagged C&I Loan Ratio	−0.002**
	(0.001)
Lagged Deposit Ratio	0.004**
	(0.002)
Lagged NPL Ratio	0.060*
	(0.036)
Fixed Effects	
Year-Quarter	Yes
Adjusted R^2	0.366
Observations	3,095

This table examines how banks' greater pre-CECL mortgage lending exposure to LMI census tracts affect allowance for loan losses on residential real estate loans. Panel A reports the summary statistics for the variables used in the analysis. Panel B reports the regression results by regressing the bank's allowance for credit losses on residential real estate loans, scaled by total assets, on its LMI Market Exposure and bank characteristics. Standard errors appear in parentheses and are clustered by banks. ***, **, and * indicate statistical significance at 1%, 5%, and 10%, respectively. All variables are defined in Appendix A.

OA.2. CECL Banks' Underwriting Standards

Table OA.2: Effect of CECL Adoption on Bank Mortgage Underwriting Standards

Panel A: Summary Statistics						
Variable	N	Mean	SD	P25	Median	P75
I(High Risk)	5,402,941	0.074	0.262	0.000	0.000	0.000
Panel B: Regression Results						
			I(High Risk)			
		(1)			(2)	
CECL × CRA × Post		−0.013***			−0.011**	
		(0.004)			(0.004)	
Log(Loan Amount)		0.012***			0.012***	
		(0.003)			(0.003)	
White		−0.019***			−0.019***	
		(0.005)			(0.005)	
Female		0.022***			0.021***	
		(0.002)			(0.002)	
Conforming		0.032***			0.033***	
		(0.004)			(0.004)	
Owner Occupied		0.080***			0.079***	
		(0.006)			(0.006)	
Loan Purpose		0.110***			0.109***	
		(0.007)			(0.007)	
Fixed Effects						
Bank-Year		Yes			Yes	
Census Tract-Year		Yes			Yes	
CECL Bank-Census Tract		Yes			Yes	
Bank-County		No			Yes	
Adjusted R^2		0.213			0.212	
Observations		4,878,824			4,878,824	

This table examines the effect of CECL adoption on bank mortgage underwriting standards. Panel A reports summary statistics for the variables used in the analysis. Panel B reports entropy-balance-weighted triple-difference estimation results of Eq. (1), in which the dependent variable is *I(High Risk)*, an indicator equals 1 if a borrower had both a high LTV ratio (i.e., above 90 percent) and a high DTI ratio (i.e., above 36 percent), for both columns. The sample consists of mortgage applications approved by banks. Standard errors appear in parentheses and are clustered by banks. ***, **, and * indicate statistical significance at 1%, 5%, and 10%, respectively. All variables are defined in Appendix A.

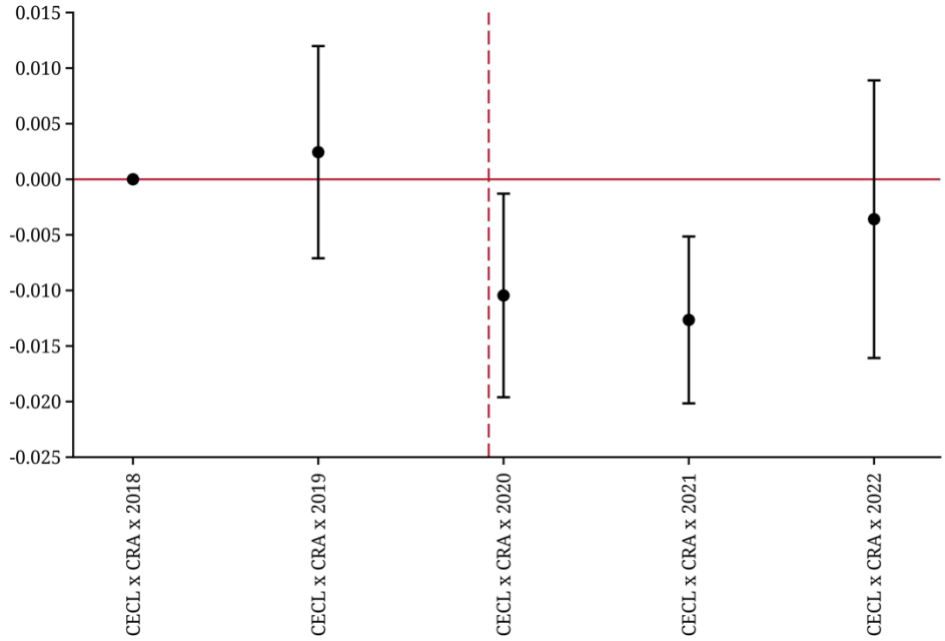


Fig. OA.2: Dynamic Effects of CECL Adoption on Bank Underwriting standards. This figure plots coefficient estimates on year-by-year indicators and their 90% confidence intervals from dynamic entropy-balanced-weighted triple-difference estimations of Eq. (1), in which dependent variables is *I(High Risk)*. The sample includes mortgage applications approved by banks. Coefficient estimates are reported relative to 2018. The regression includes bank-year, census tract-year, CECL bank-census tract, and bank-county fixed effects, and all borrower and loan characteristics control variables. Standard errors are clustered by banks. All variables are defined in Appendix A.

OA.3. Placebo Tests for Bank Mortgage Lending

I first examine federally insured or guaranteed mortgages. Federal insurance and guarantees absorb a substantial portion of credit losses on FHA and FSA/RHS loans. FHA insurance covers 100% of the outstanding loan balance, the USDA Rural Housing Service's Section 502 program provides lenders with a 90% loan-note guarantee, and Farm Service Agency guarantees may cover up to 95% of losses on principal and interest. I re-estimate the entropy-balanced model specified in Eq. (1) using these mortgages. Panel A of Table OA.3 in Online Appendix OA.3 reports economically small and statistically insignificant effects on approvals and interest rates.

I then examine lending under the SBA 7(a) program, in which the SBA guarantees up to 85 percent of the loan amount for loans of $150,000 or less and up to 75 percent for loans above $150,000. The SBA data, however, report the borrower's ZIP code rather than its census tract. I construct a continuous measure of each ZIP code's exposure to LMI census tracts, *ZIP LMI Share*. For each year-quarter, I use the HUD–USPS ZIP-to-tract crosswalk to identify all census tracts associated with a ZIP code and merge the corresponding tract-level LMI classifications. As the crosswalk reports the proportion of a ZIP code's business addresses located in each census tract, I then calculate the ZIP-level LMI share as the business-address-weighted average of the tract-level LMI indicators.

I aggregate the loan-level SBA data to the bank–ZIP code–quarter level and define the dependent variable as the natural log of total SBA loan amount originated by a bank in a ZIP code during the quarter. I then estimate an entropy-balance-weighted triple-difference specification, replacing *CRA* with the continuous ZIP-level LMI share. Panel B of Table OA.3 shows no evidence that CECL banks reduced SBA lending disproportionately in ZIP codes with greater exposure to LMI census tracts after CECL adoption.

Table OA.3: Placebo Tests Using Government-Insured and Government-Guaranteed Loans

Panel A: FHA-Insured and RHS/FSA-Guaranteed Mortgage Loans		
	I(Approved)	Interest Rate
	(1)	(2)
CECL × CRA × Post	0.011	0.003
	(0.015)	(0.040)
Control		
All Loan and Borrower Characteristics	Yes	Yes
Fixed Effects		
Bank-Year	Yes	Yes
Census Tract-Year	Yes	Yes
CECL Bank-Census Tract	Yes	Yes
Bank-County	Yes	Yes
Adjusted R^2	0.157	0.592
Observations	532,351	448,203
Panel B: Small Business Administration Loans		
	Log(Loan Amount)	
	(1)	
CECL × ZIP LMI Share × Post	−0.033	
	(0.216)	
Fixed Effects		
Bank-Year-Quarter	Yes	
ZIP Code-Year-Quarter	Yes	
CECL Bank-ZIP Code	Yes	
Bank-County	Yes	
R^2	0.881	
Within R^2	0.001	
Observations	102,083	

This table reports entropy-balance-weighted triple-difference estimates for government-insured or government-guaranteed loans. Panel A uses mortgage applications insured by the Federal Housing Administration (FHA) or guaranteed by the Rural Housing Service or Farm Service Agency (RHS/FSA) to estimate Eq. (1). Dependent variables are *I(Approved)* in Column (1) and *Interest Rate* in Column (2). Panel B estimates a modified version of Eq. (1) using Small Business Administration (SBA) loans. The main independent variable is *CECL × ZIP LMI Share × Post,* and the dependent variable is *Log(Loan Amount)*, measured at the bank-ZIP code-quarter level. Standard errors appear in parentheses and are clustered by banks. ***, **, and * indicate statistical significance at 1%, 5%, and 10%, respectively. All variables are defined in Appendix A.

OA.4. Corroborating Analysis using Banks Adopting CECL in 2023

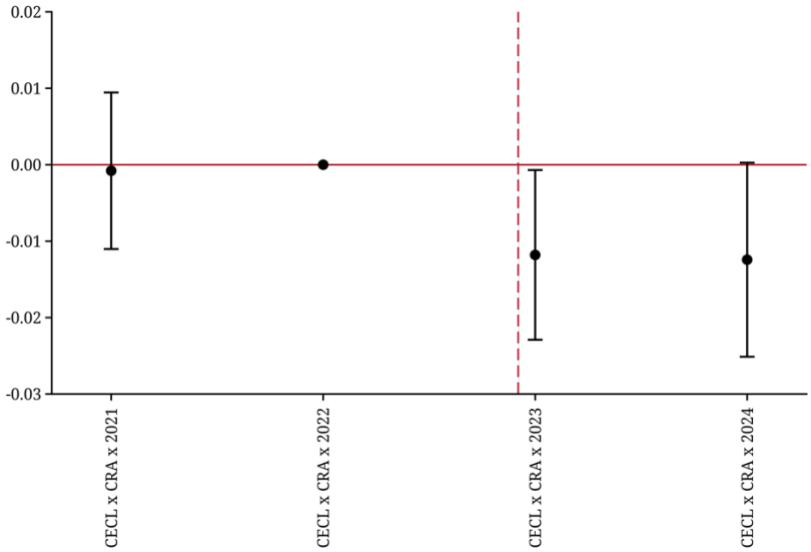
Table OA.4: Entropy Balancing Statistics

Panel A: Summary Statistics of Bank Characteristics						
	N	Mean	SD	p25	Median	p75
Log(Size)	710	23.640	1.199	22.696	23.324	24.255
Deposit Ratio	710	0.824	0.054	0.795	0.830	0.861
Tier-1 Capital Ratio	710	0.093	0.015	0.083	0.091	0.099
Residential R/E Loan Ratio	710	0.167	0.114	0.094	0.140	0.212
C&I Loan Ratio	710	0.117	0.070	0.066	0.109	0.159
NPL Ratio	710	0.004	0.004	0.002	0.003	0.005

Panel B: Entropy Balancing Statistics of Bank Characteristics			
	Treatment	Control	
	Mean	Pre-balancing Mean	Post-balancing Mean
Log(Size)	23.135	23.841***	23.135
Deposit Ratio	0.827	0.823	0.827
Tier-1 Capital Ratio	0.094	0.092	0.094
Residential R/E Loan Ratio	0.186	0.160**	0.186
C&I Loan Ratio	0.098	0.124***	0.098
NPL Ratio	0.004	0.004	0.004

This table reports summary and entropy balancing statistics of bank characteristics for the corroborating analysis using banks adopting CECL in 2023. The treatment and control banks are balanced by means on characteristics, including *Log(Size)*, *Tier-1 Capital Ratio*, *Deposit Ratio*, *Residential R/E Loan Ratio*, *C&I Loan Ratio*, and *NPL Ratio*. Panel A reports summary statistics, and Panel B reports entropy balancing statistics, including means for the control banks both before and after applying entropy balancing weights. ***, **, and * represent statistical significance at the 1%, 5%, and 10% levels, respectively, for tests of differences between the means of characteristics of the treatment and control banks. Continuous variables are winsorized at the 1% level. All variables are defined in Appendix A.

Panel A: I(Approved)



Panel B: Interest Rate

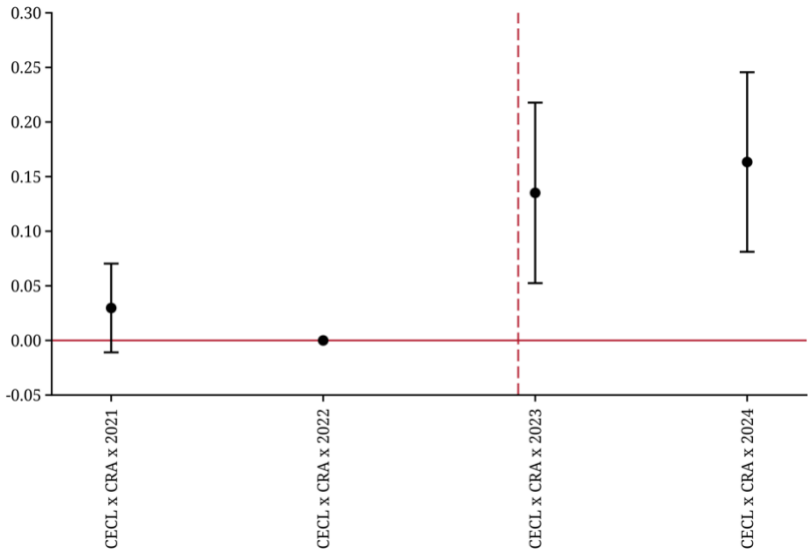


Fig. OA.4: Dynamic Effects of CECL Adoption in 2023. This figure plots coefficient estimates on year-by-year indicators and their 90% confidence intervals from dynamic entropy-balance-weighted triple-difference estimations for banks' CECL adoption in 2023. Panel A reports estimates for *I(Approved)*, and Panel B reports estimates for *Interest Rate*. The sample in Panel A includes mortgage applications, and the sample in Panel B includes originated mortgages. Coefficient estimates are reported relative to 2022, the last pre-adoption year. The regressions include bank-year, census tract-year, treatment bank-census tract, and bank-county fixed effects, as well as all borrower and loan characteristics control variables. Standard errors are clustered by banks. All variables are defined in Appendix A.

OA.5. Geographic Comparability

Table OA.5: Geographic Overlap between CECL and Non-CECL Banks

	I(Approved)	Interest Rate
	(1)	(2)
Panel A: Counties with Both CECL and Non-CECL Banks		
CECL × CRA × Post	−0.014***	0.030**
	(0.004)	(0.013)
Control		
Loan and Borrower Characteristics	Yes	Yes
Fixed Effects		
Bank-Year	Yes	Yes
Census Tract-Year	Yes	Yes
CECL Bank-Census Tract	Yes	Yes
Bank-County	Yes	Yes
Adjusted R^2	0.271	0.626
Observations	5,386,257	4,721,405
Panel B: Census Tracts with Both CECL and Non-CECL Banks		
CECL × CRA × Post	−0.014***	0.030**
	(0.004)	(0.013)
Control		
Loan and Borrower Characteristics	Yes	Yes
Fixed Effects		
Bank-Year	Yes	Yes
Census Tract-Year	Yes	Yes
CECL Bank-Census Tract	Yes	Yes
Bank-County	Yes	Yes
Adjusted R^2	0.264	0.625
Observations	5,170,203	4,540,464

This table reports entropy-balance-weighted triple-difference estimation results of Eq. (1) requiring CECL and non-CECL banks operate in the same geographies. Panel A restricts the sample to counties in which both CECL and non-CECL banks originate mortgages. Panel B retains only census tracts in which both CECL and non-CECL banks originate mortgages. The dependent variables are *I(Approved)* in Column (1) and *Interest Rate* in Column (2). The sample in Column (1) consists of mortgage applications, whereas the sample in Column (2) consists of originated mortgages. Standard errors appear in parentheses and are clustered by banks. ***, **, and * indicate statistical significance at 1%, 5%, and 10%, respectively. All variables are defined in Appendix A.

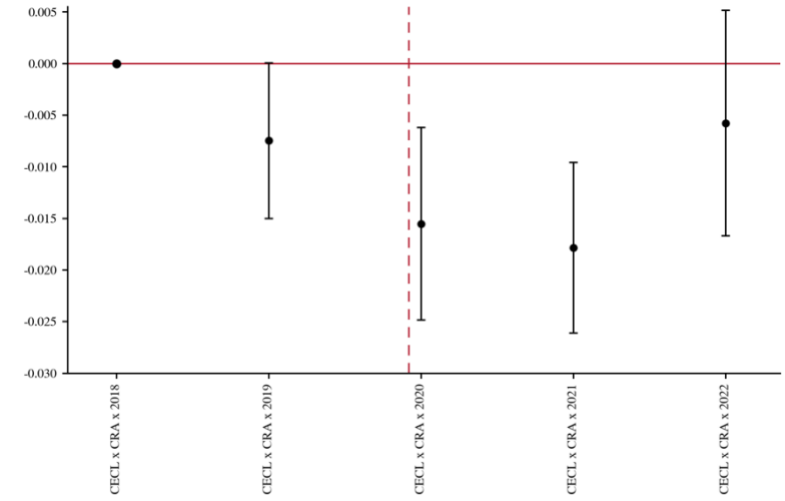
OA.6. Alternative Size Threshold

Table OA.6: Alternative Size Threshold

	I(Approved)	Interest Rate
	(1)	(2)
CECL × CRA × Post	−0.011***	0.037***
	(0.004)	(0.013)
Controls		
Loan and Borrower Characteristics	Yes	Yes
Fixed Effects		
Bank × Year	Yes	Yes
Census Tract × Year	Yes	Yes
CECL Bank × Census Tract	Yes	Yes
Bank × County	Yes	Yes
Entropy-Balancing Weights	Yes	Yes
Adjusted R^2	0.275	0.637
Within R^2	0.060	0.098
Observations	5,909,609	5,139,676

This table reports entropy-balance-weighted triple-difference estimation results of Eq. (1) using a maximum bank size threshold of $50 billion. The dependent variables are *I(Approved)* in Column (1) and *Interest Rate* in Column (2). The sample includes mortgage applications submitted to CECL and non-CECL banks from 2018 through 2022. Standard errors appear in parentheses and are clustered by banks. ***, **, and * indicate statistical significance at 1%, 5%, and 10%, respectively. All variables are defined in Appendix A.

Panel A: I(Approved)



Panel B: Interest Rate

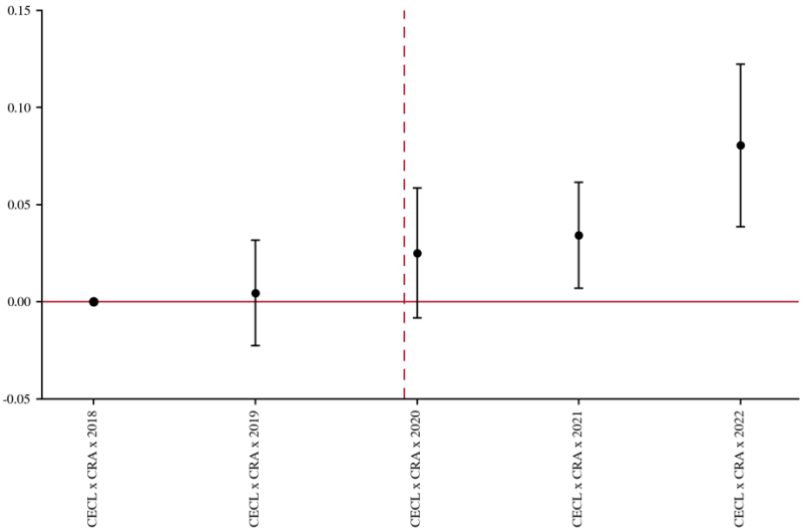


Fig. OA.6: Dynamic Effects of CECL Adoption Using a $50 Billion Bank Size Threshold. This figure plots coefficient estimates on year-by-year indicators and their 90% confidence intervals from dynamic entropy-balanced-weighted triple-difference estimations of Equation (1). The dependent variables are *I(Approved)* and *Interest Rate*. The sample includes mortgage applications submitted to banks in Panel A and mortgages originated by banks in Panel B. The maximum bank size threshold is increased from $30 billion to $50 billion. Coefficient estimates are reported relative to 2018. The regressions include bank–year, census tract–year, CECL bank–census tract, and bank–county fixed effects, and borrower and loan characteristics control variables. Standard errors appear in parentheses and are clustered by banks. ***, **, and * indicate statistical significance at 1%, 5%, and 10%, respectively. All variables are defined in Appendix A.

OA.7. Non-CECL Banks' Mortgage Credit Supply in LMI Census Tracts

To examine whether non-CECL banks increased their supply of mortgage credit in LMI census tracts, I use the following DiD estimator:

$$I(Approved)_{i,b,c,s,t} = \beta_1 (CRA_c \times Post_t) + \Gamma' X_{i,t} + \alpha_{b,t} + \delta_c + \lambda_{s,t} + \mu_{b,s} + \varepsilon_{i,b,c,s,t}$$

where i indexes mortgage applications, b indexes banks, c indexes census tracts, s indexes counties, and t indexes years. The vector $X_{i,b,c,s,t}$ contains loan and borrower characteristics. The specification includes bank–year, census tract, county–year, and bank–county fixed effects. Column (1) of Table OA.7 reports results, with no significant change in non-CECL banks' mortgage credit supply in LMI census tracts. To address the concern that divergent economic trends between LMI and non-LMI census tracts bias coefficient estimates, I further restrict analyses to mortgage applications from non-CECL banks in census tracts with an MFI between 70 and 90 percent of the surrounding area's MFI. Column (2) of Table OA.7 reports results, with no significant change in non-CECL banks' mortgage credit supply in these LMI census tracts.

I further test whether non-CECL banks increased mortgage credit supply in LMI census tracts that are exposed more to CECL banks, using mortgage applications submitted to non-CECL banks in LMI census tracts, by estimating Eq. (2). Column (3) of Table OA.7 reports results. Non-CECL banks did not increase mortgage credit supply in LMI census tracts that are exposed more to CECL banks.

Table OA.7: Mortgage Approval of Non-CECL Banks in LMI Census Tracts

	I(Approved)		
	Full Sample	MFI [70%, 90%]	CECL Exposure
	(1)	(2)	(3)
CRA × Post	0.003	−0.003	0.001
	(0.002)	(0.002)	(0.004)
CECL Share × Post			0.001
			(0.015)
CECL Share × CRA × Post			0.000
			(0.013)
Control			
Loan and Borrower Characteristics	Yes	Yes	Yes
Fixed Effects			
Bank-Year	Yes	Yes	Yes
Census Tract	Yes	Yes	Yes
County-Year	Yes	Yes	Yes
Bank-County	Yes	Yes	Yes
Adjusted R^2	0.164	0.175	0.157
Observations	3,417,024	504,603	3,242,587

Column (1) reports the DiD estimation results using mortgage applications submitted to non-CECL banks. Column (2) reports the DiD estimation results using mortgage applications submitted to non-CECL banks in census tracts with an MFI between 70% and 90% of the surrounding area's MFI. Column (3) reports the estimation results of Eq. (2) using mortgage applications submitted to non-CECL banks. The dependent variable is *I(Approved)* in all columns. Standard errors appear in parentheses and are clustered by banks. ***, **, and * indicate statistical significance at 1%, 5%, and 10%, respectively. All variables are defined in Appendix A.

OA.8. Placebo Test for Non-bank Lenders

Table OA.8: Placebo Test for Non-bank Lenders Using Government-Backed Mortgage Loans

	I(Approved)	Interest Rate
	(1)	(2)
CECL Share × CRA × Post	0.002	0.012
	(0.009)	(0.016)
Control		
Loan and Borrower Characteristics	Yes	Yes
Fixed Effects		
Lender-Year	Yes	Yes
County-Year	Yes	Yes
Census Tract	Yes	Yes
Lender-County	Yes	Yes
Adjusted R^2	0.202	0.643
Observations	3,993,387	3,201,749

This table reports entropy-balance-weighted triple-difference estimates of Eq. (2) using mortgage applications insured by the Federal Housing Administration (FHA) or guaranteed by the Rural Housing Service or Farm Service Agency (RHS/FSA) submitted to non-bank lenders. The dependent variable is *I(Approved)* in Column (1) and *Interest Rate* in Column (2). Standard errors appear in parentheses and are clustered by lenders. ***, **, and * indicate statistical significance at 1%, 5%, and 10%, respectively. All variables are defined in Appendix A.